



OPEN Dynamic appliance scheduling and energy management in smart homes using adaptive reinforcement learning techniques

Poonam Saroha¹, Gopal Singh¹, Umesh Kumar Lilhore², Sarita Simaiya², Monish Khan³✉, Roobaea Alroobaea⁴, Majed Alsafyani⁴ & Hamed Alsufyani⁵

Smart home energy management is complicated because of varying user preferences, expenses, and consumption. These dynamics are difficult for traditional systems to handle, but new developments in reinforcement learning and optimization may be able to help. The paper introduces a novel Demand Response (DR) method that integrates a Self-Adaptive Puma Optimizer Algorithm (SAPOA) with a Multi-Objective Deep Q-Network (MO-DQN), improving smart home energy consumption, cost, and user preferences management. SAPOA adaptively maximizes numerous objectives, while DQN improves decision-making by assimilating interactions. The proposed method adapts to user preferences by learning from previous energy usage patterns and optimizing the scheduling of critical household appliances, enhancing energy efficiency. Static optimization in traditional home energy management systems (HEMS) makes it difficult to handle changing expenses and dynamic user preferences. Reinforcement learning (RL) methods now in use frequently lack sophisticated optimization integration. The experimental results show that the outperforming multiobjective reinforcement learning puma optimizer algorithm (MORL-POA), SAPOA, and POA methods, the suggested solution dramatically lowers the peak-to-average ratio (PAR) value from 3.4286 to 1.9765 without RES and 1.0339 with RES. By combining SAPOA with DQN, the suggested approach maximizes energy management, optimizes appliance scheduling, and efficiently manages uncertainty, improving performance and flexibility. Metrics like peak average ratio (PAR), energy usage, and electricity cost are used to assess performance, while the Matlab platform is used for implementation.

Keywords Demand response, Deep Q network, Self-adaptive Puma optimizer algorithm, Energy consumption, Appliance scheduling

Smart HEMS are a necessity of modern homes, controlling appliances automatically and adjusting settings for efficient and sustainable energy consumption¹. Smart HEMS employ IoT, AI, and machine learning to monitor and control electricity usage in real-time². Smart meters and intelligent controllers, with sensors, smart HEMS allocate power efficiently and program appliances to operate at the appropriate times. Smart HEMS apply dynamic pricing to minimize energy spending and enhance sustainability by controlling energy supply and demand, responding to price fluctuation, and considering user preferences^{3,4}.

The importance of smart HEMS grows as energy consumption becomes more complex. These systems increase energy efficiency and integrate renewable energy systems to help store energy and reduce environmental degradation⁵. Demand Response (DR) technology controls energy usage as a function of price volatility or peak load, improving grid stability, reducing energy costs, and enabling renewable energy integration^{6,7}. DR programs, through time-of-use rates, automated devices, and monetary incentives, induce users to change their electricity consumption patterns, leading to more efficient and reliable energy management⁸.

Smart HEMS employ advanced technologies to deliver maximum performance in a range of real-world applications⁹. Through the application of adaptive algorithms and real-time information, dynamic appliance

¹Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak, Haryana, India. ²School of Computing Science and Engineering, Galgotias University, Greater Noida, UP, India. ³Arba Minch University, Arba, Minch, Ethiopia. ⁴Department of Computer Science, College of Computers and Information Technology, Taif University, P. O. Box 11099, Taif 21944, Saudi Arabia. ⁵Department of Computer Science, College of Computing and Informatics, Saudi Electronic University, Riyadh 11673, Saudi Arabia. ✉email: drkumacse@gmail.com

scheduling optimizes the efficiency of home appliances, leading to a significant reduction in energy consumption and enhanced user comfort^{10,11}. To maximize utilization of renewable energy resources and minimize grid dependency, smart systems manage distribution and storage of energy as well as photovoltaic panels and storage batteries¹². Load forecasting applications and DR help in predicting future energy demand and shifting consumption to off-peak hours, cutting costs and system load¹³. They can manage their use comfortably via web pages or mobile apps with remote control and monitoring¹⁴.

However, traditional methods such as heuristic techniques, rule-based systems, and basic reinforcement learning algorithms are inflexible and insensitive to dynamic changes in real-time energy usage^{15,16}. Such methods provide suboptimal solutions and are incapable of satisfactorily managing the multi-objective nature of energy management^{17,18}. For example, rule-based systems cannot respond to dynamic energy prices and user preferences, resulting in inefficient energy consumption.

The proposed work presents a new smart HEMS by incorporating a SAPOA and a MO-DQN. This framework adaptively modifies according to users' needs, minimizes the cost even without the availability of renewables, and enhances efficiency. The integration is novel since it merges the best of adaptive metaheuristic optimization and reinforcement learning to enable the system to learn from the history of usage and optimize appliance scheduling better than traditional methods. The SAPOA adds the flexibility of the optimization process, and the MO-DQN provides a means to balance various objectives like cost savings and user comfort.

The major contributions of the paper are as follows:

- Introduced a new DR approach integrating a SAPOA with a Multi-Objective DQN, able to handle in a much more effective manner the dynamic energy consumption, cost, and user preferences.
- The proposed approach changes to changing user preferences by learning from previous usage, and optimizing the scheduling of important household appliances like air conditioners, washing machines, and refrigerators, thus improving energy efficiency and lowering expenses.

The sections that follow are organized as follows: In part 2, applicable research and literature reviews are examined; in part 3, the proposed framework is introduced; in part 4, the observed results and discussions are carefully analyzed; in part 5, the study's final evaluation is provided and in part 6 future scope about this research work is discussed.

Literature review

Smart home energy management is now a significant area of research since the need for green and efficient energy solutions continues to rise. With an explosion of smart appliances, devices, and renewable resources, intelligent systems that can manage energy usage dynamically while reducing expenditure and preserving comfort are in higher demand. Reinforcement learning (RL) has become a potential answer to these challenges because it can learn by interaction, accommodate user preferences, and manage the variability of dynamic and uncertain environments.

One such significant contribution is the use of multi-objective reinforcement learning (MORL) in smart homes to maximize different parameters such as cost, power consumption, and user comfort. Afroosheh et al. (2024) and Pinthurat et al. (2024) both highlight the application of advanced RL methods, such as Q-learning and deep reinforcement learning (DRL), to schedule home appliances dynamically based on user requirements, environmental conditions, and current energy prices. To provide the best performance in different situations, these strategies tend to strike a balance between user comfort and energy efficiency. For instance, with AI-based RL, Afroosheh et al. (2024) propose an optimal energy management system based on layout that enhances system flexibility and adapts to user requirements. By considering the uncertainty of household parameters and external energy supply, the integration of energy storage and renewable energy sources, as shown in the research of Ren et al. (2024), further improves the optimization process.

The studies under review utilize a range of RL algorithms and frameworks in their methodology. As an example, Abid et al. (2024) apply a multi-agent deep reinforcement learning method for microgrid resource planning multi-objective optimization. To optimize energy management in residential areas with many appliances and energy sources, this method facilitates decentralized decision-making. Likewise, through the combination of hybrid evolutionary algorithms and RL methodologies for sustainable energy management of buildings, Li and Xu (2024) illustrate how malleable RL can be across numerous scenarios. These studies utilize most of them with simulated datasets, and extremely few provide real data or experiments on real smart home settings. This restricts the external validity of the results but emphasizes the necessity for additional research that assesses the scalability and robustness of RL-based systems with actual data.

The main conclusions of these studies emphasize the significant cost reduction and energy management improvements. Ren et al. (2024), for instance, demonstrate how their RL-based optimization framework reduces domestic energy expenses by accounting for uncertain factors such as domestic usage trends and weather. Similarly, Abishu et al. (2024) highlight how multi-agent RL can be used to maximize demand response (DR) for real-time energy management, reducing peaks in energy usage and enhancing grid stability. These results illustrate that by considering user preferences and requirements, RL can preserve or enhance user satisfaction while significantly lowering energy usage and electricity bills. As the advantages of RL-based systems in smart homes are evident, these studies also point out the disadvantages, and most notably concerning model complexity and computational requirements. Common limitations involve uncertain energy costs, high-quality data requirements, and the challenge of training RL models with multiple user preferences. Moreover, reliance on simulations over real-world testing limits the findings' generalizability to practical smart home setups.

Alatawi (2024) proposed a hybrid model for the optimization of HEMS through Bacterial Forging Metaheuristic Optimization and Deep Reinforcement Learning. The given research improved energy consumption efficiency,

reduced spending, and implemented renewable energy in smart homes. Nonetheless, computational intricacy, scalability issues, and dealing with vague consumption patterns and momentary user preferences made it difficult for the model. These disadvantages highlight the need for stronger and more adaptable answers in smart home energy management. In another field, Aldahmashi and Ma (2024) applied machine learning and deep learning techniques to derive deceptive data and sentiment trends from user-content such as product reviews and news articles. The analysis employed the OSEMN framework and DeLTran15 model for classifying humanitarian posts from the X platform with greater efficiency and interpretability. However, the model had some drawbacks, including misclassification, challenges in dealing with imbalanced data, and potential overfitting due to the small training dataset. These challenges highlight the importance of good-quality data and robust models in achieving reliable and accurate results.

Relating these conclusions to the context of HEMS in general, these systems collect information and regulate appliances for demand-side management. They can use classical models or data-driven methods like RL, which require load and generation of data and significant computational resources. A new HEMS is proposed by Real et al.¹⁹, for optimizing electric battery operation using advanced load forecasting techniques in 2024. Various RL agents are qualified with different algorithms to reduce electricity costs and increase self-consumption in a residential system with a PV battery. The incorporation of renewable energy sources in smart grids is a promising solution.

However, optimizing hybrid systems is a key research area. Sankarananth et al., 2023 proposed a method using AI and metaheuristic algorithms for handling renewable energy in smart grids. CNN-PSO algorithm is the most successful in forecasting renewable energy generation. These findings advance hybrid systems in smart grids, enhancing energy production and distribution. Cortez et al. 2023 assessed Intraday Block Pricing (IBP) for power system demand management. They utilized a bilevel optimization problem to reduce PAR and promote DR. Simulation studies indicated the proposed IBP decreased PAR and was an effective mechanism of fair pricing among residences. Nevertheless, the research was limited by factors such as simplistic assumptions, non-convexity, non-scalability, restriction in applicability, and inadequate proof of validation. More research with empirical testing is advisable.

The integration of RL into energy management for smart homes has exhibited tremendous potential with regard to energy efficiency, cost reduction, and user satisfaction. While the methods presented in the papers under review are substantial leaps forward, data quality, computational complexity, and practical implementation issues remain challenges. Future work in this field needs to be aimed at enhancing the scalability of RL models, applying them to real-world scenarios, and further tuning algorithms to address the intricacies of dynamic energy systems and user behavior. Table 1 shows some recent research into HEMS.

Proposed methodology

The suggested model seeks to improve smart home energy management through optimized energy use, cost minimization, and enhanced user satisfaction. It uses real-time energy prices to schedule appliances for optimization, taking into account user preferences. The system is dynamic and adaptive, learning from previous usage patterns. It targets large energy-using appliances and optimizes efficiency through sophisticated algorithms such as the SAPOA and MO-DQN.

Reference	Key Method	Contributions	Outcomes	Future Directions
Pinthurat et al. (2024)	RL-based energy management with storage	Overview of RL for smart homes with energy storage optimization.	Enhanced understanding of RL for energy storage.	Focus on real-time management and renewable integration.
Afroosheh et al. (2024)	Layout-based RL for energy optimization	AI-driven RL approach for optimal appliance scheduling in smart homes.	Cost reduction and improved scheduling flexibility.	Real-time applications and dynamic energy demand testing.
Ren et al. (2024)	Data-driven DRL for uncertain parameters	Framework for optimizing energy management with uncertain household factors.	Increased energy efficiency under uncertainties.	Integration of real-time data and robustness enhancement.
Abid et al. (2024)	Multi-agent DRL for microgrid planning	Multi-agent DRL for resource allocation in microgrids.	Improved resource management and energy efficiency.	Scalability for larger systems and diverse energy sources.
Abishu et al. (2024)	Multi-agent DRL for demand response	Optimizing demand response using multi-agent DRL in real-time.	Optimized energy consumption and reduced peak demand.	Real-world data integration for diverse consumption patterns.
Li and Xu (2024)	Hybrid evolutionary and ML for buildings	ML and evolutionary algorithms for smart building energy optimization.	Flexible energy optimization based on building type.	Integration with energy storage and user behavior adaptation.
Islam et al. (2024)	ML for appliance energy optimization	Predictive energy optimization for appliances using historical data.	Reduced consumption with better scheduling.	Enhancing predictions with real-time data and automation integration.
Kanso et al. (2024)	Automated energy management framework	AI-based automated energy management system for smart homes.	Increased energy efficiency through automation.	Real-time adaptability and renewable integration.
Yang et al. (2024)	Optimal attitude control for satellites	Optimal control for satellites' energy management (not relevant to smart homes).	Insights into satellite energy control.	Further control methods research for renewable energy systems.
Saberi et al. (2024)	Multi-agent Q-learning for building management	Multi-agent Q-learning for smart building energy management under uncertainty.	Effective energy management in uncertain conditions.	Scaling to larger networks and energy source diversity.
Raza et al. (2024)	Survey on smart home energy systems	Comprehensive review of challenges and solutions in smart home energy systems.	Insights into research gaps and solutions in the field.	Calls for interdisciplinary research and standardization.
Kaewdornhan et al. (2023)	Multi-agent DRL for multi-home energy management	Multi-agent DRL for multi-home energy management, including EV scheduling.	Optimized energy consumption and EV charging.	Dynamic scheduling and vehicle-to-grid (V2G) systems integration.

Table 1. Research gaps from the existing works conciseness:

The model gathers information on energy usage, user preferences, and cost variations to guide decisions. It examines user preferences and employs adaptive learning to enhance scheduling effectiveness. The system maximizes appliance operation, minimizing energy wastage and advancing smart HEMS. Performance metrics like PAR and energy consumption levels are used to evaluate the effectiveness of energy management strategies. The system incorporates feedback loops for continuous refinement and responds to changes in energy costs and user preferences, ensuring robust performance and user-centricity. The overall architecture of the proposed methodology is shown in Fig. 1.

Models of systems and formulation of problems

The main electricity consumers, such as refrigerators, washers, and air conditioners, are the focus of the smart HEMS. This is how it improves energy efficiency and reduces costs through advanced scheduling and learning mechanisms for the optimization of operations. The system will have a comprehensive and effective strategy for managing household energy consumption in a very dynamic and efficient way.

- Refrigerator: The Energy Consumption of the refrigerator appliance is described is expressed in Eq. (1)

$$Energy^1 = \sum_{r=1}^R \sum_{m=1}^{M_1} Energy_{m,r}^1 \quad (1)$$

Refrigerators require constant operation and cannot have their energy consumption changed. $Energy_{m,r}^1$ characterizes energy consumption of a time slot R , refrigerators m , where $m \in \{1, 2, 3, \dots, M_1\}$ and $r \in \{1, 2, 3, \dots, R\}$. Here, M_1 and R stand for the optimization window and the number of rigid appliances, respectively²⁰. The total energy used by all of the rigid appliances is provided by Eq. (1)

- Washing machines: Users can designate an operation period and submit the request to the EMS. During this operating period, the EMS has to complete the task. Washing machine appliances m in time slot R have

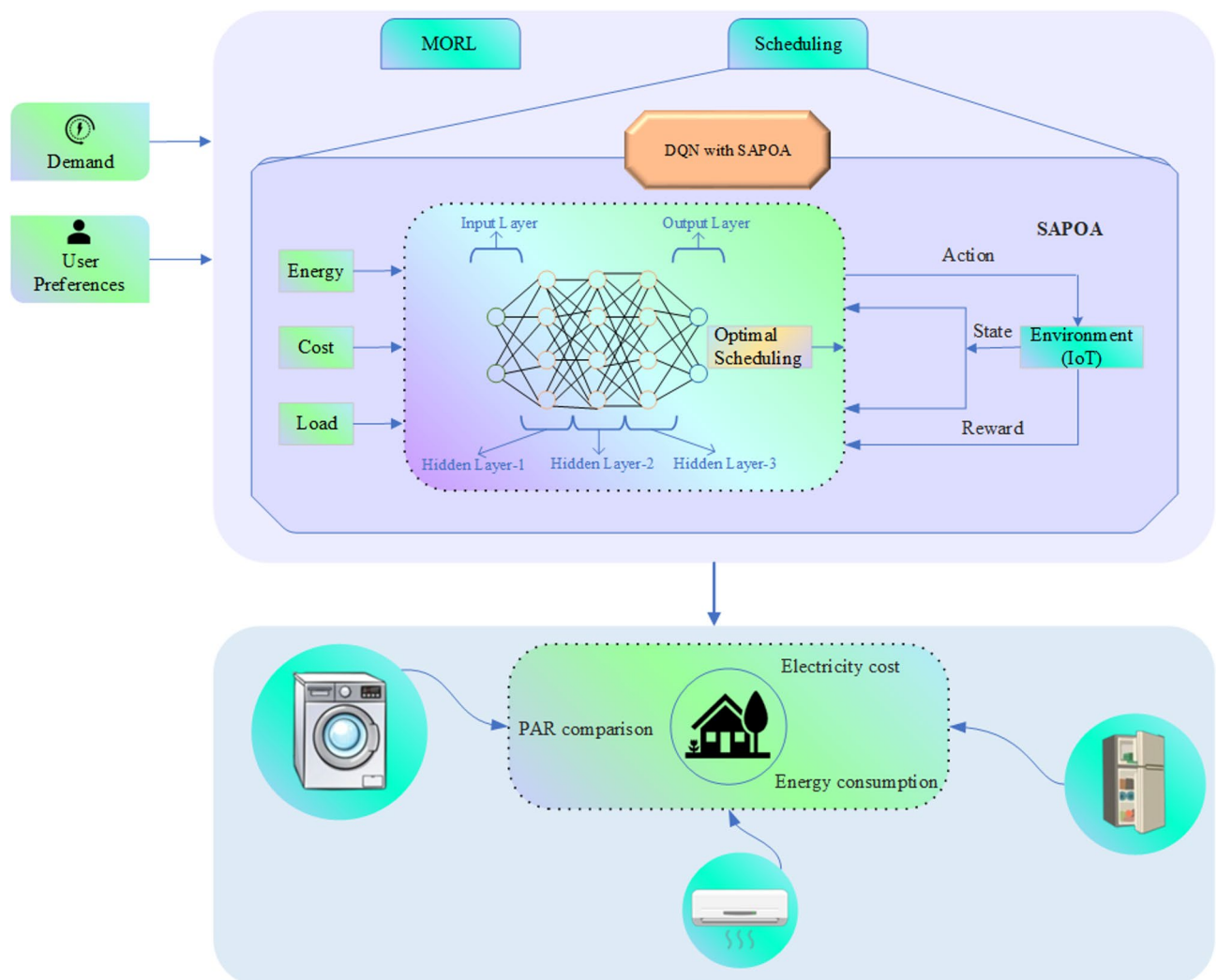


Fig. 1. Architecture of the Home Energy Management System.

energy consumptions denoted as $Energy_{m,r}^2$, where $m \in \{1, 2, 3, \dots, M_2\}$, where M_2 is the number of washing machine appliances. Throughout the observation period, washing machine appliances used the following amounts of energy²¹ as described in Eq. (2)

$$Energy^2 = \sum_{r=1}^R \sum_{m=1}^{M_2} Energy_{m,r}^2 \quad (2)$$

Shifting device operation time can lead to user dissatisfaction. User dissatisfaction increases as operation time is shifted. EMS can adjust the start time within the operation period. Different appliances have different priority levels. θ_m^2 represents appliance dissatisfaction parameter. This parameter captures user dissatisfaction with energy consumption levels in smart HEMS. It is measured through a formula that compares energy usage to predefined thresholds, with increased consumption reflecting increased dissatisfaction. The parameter is affected by appliance priority levels, with higher-priority appliances contributing more to dissatisfaction. Validation is conducted through empirical testing, user questionnaires, and statistical analysis to match dissatisfaction levels to user input and usage data. This ensures effective alignment of the system with the user's preference. As higher θ_m^2 means higher priority, user dissatisfaction is linked to operation time and parameters. User discontent $User_{m,r}^2$ ($\theta_m^2, r_{m,st}^2$) is represented by the variable restrictions in Eq. (3)

$$r_{m,st}^2 < r_{m,rt}^2 < r_{m,et}^2 \quad (3)$$

Where, $r_{m,st}^2$ is the start time, $r_{m,rt}^2$ is the request time, $r_{m,et}^2$ is the end time of operation carried out by EMS. User dissatisfaction arises from the fact that appliance operations fail to comply with user preferences, as observed in Eq. (3). To ensure user satisfaction, the EMS must schedule appliance start and end times before the requested time. Failure to meet these requirements brings about user discontent due to undesirable operational modifications²².

- Air conditioner: Air conditioners with variable power consumption capabilities can function within the range of minimal and maximum energy requirements. $Energy_{m,r}^3$, where $m \in \{1, 2, 3, \dots, M_3\}$ and M_3 is no. of Air conditioner appliances, represents energy consumption of Air conditioner m in time slot R . All Air conditioner applications' energy usage is determined in Eq. (4)

$$Energy^3 = \sum_{r=1}^R \sum_{m=1}^{M_3} Energy_{m,r}^3 \quad (4)$$

The energy demand for an Air conditioner appliance is shown as $D_{m,max}^3$, with $D_{m,min}^3$ for minimum demand. Energy usage is controlled by EMS within the range of $D_{m,min}^3$ to $D_{m,max}^3$. θ_m^3 indicates the unhappiness parameter for the appliance m , with $0 < \theta_m^3 < 1$, implying higher priority with a larger θ_m^3 ²³. User discontent with the appliance is affected by energy consumption and dissatisfaction, with specific variable constraints described in Eq. (5).

$$D_{m,min}^3 < Energy_{m,r}^3 < D_{m,max}^3 \quad (5)$$

From Eq. (5) user discontent can be calculated as a function of energy consumption and unhappiness parameters. Higher unhappiness parameter values indicate greater dissatisfaction. If energy consumption is at its minimum, the discontent level is zero, indicating no discontent. Conversely, if energy consumption is at its maximum, the discontent level is θ_m^3 , reflecting maximum discontent based on user priorities²⁴.

DR approach based on a multi-objective improved reinforcement learning (MORL)

DR denotes the strategy with a set of technologies and programs intended to manage consumer demand for electricity. DR works by encouraging consumers to alter their consumption patterns in response to electricity price fluctuations or incentives offered for reducing their energy use during peak periods, to ensure stability of grid operation, peaking demand reduction, and better integration of renewable energies. Crucial elements of the DR include time-based rates; real-time pricing, critical peak pricing, and time-of-use²⁵.

In these charges, the electricity price varies during the day following the actual variation of generation and delivery costs. Incentive-based programs provide monetary incentives to the consumer who reduces his consumption during peak periods or shifts load to off-peak times. Examples of these programs include Interruptible Load programs and Direct Load Control. Automated DR will utilize sophisticated technology, smart meters, sensors, and automated control systems in near real-time to monitor and execute the DR signal in an automated response. DR will maintain grid reliability by balancing supply and demand to prevent shutdowns and ensure grid stability¹⁹. Due to the unpredictability of energy prices, renewable energy sources, and potential changes in user preferences, this section presents a user preference-based DR strategy, that uses a MORL algorithm. The home energy management system is a challenging process owing to the dynamic nature of cost, energy consumption, and user preference. The HEMS controls appliance functioning to lower energy costs and user dissatisfaction. The basis of MORL is a Markov decision process²⁶.

In this case, $State_m$ represents the state during the time slot R , where $state_{m,r} \in State_m$, and $State_m$ is the state set of agent m . Agent m given $State_m$, chooses an action $action_{m,r}$ from its action set $Action_m$,

moves on to the subsequent state $State_{m, r+1}$, and is rewarded with $Reward_{m, r+1}$ by the environment. Action $action_{m, r}$ assessed using the reward $Reward_{m, r+1}$. Finding an ideal policy, or an ideal mapping from states to actions, maximizes the expected value of the cumulative prize, is an agent. m 's primary objective. The action value of the pair ($state_{m, r}$; $action_{m, r}$) is defined as follows in Eq. (6): given a $state_{m, r}$, an action $Action_{m, r}$, and a policy π_m^* .

$$Op_{m, \pi_m}(state_{m, r}, action_{m, r}) = E_{\pi_m} \left[\sum_{r=1}^R \gamma^{r-1} reward_{m, r} | state_{m, r}, action_{m, r} \right],$$

$$\forall state_{m, r} \in State_m, \forall action_{m, r} \in Action_m \quad (6)$$

Where γ represents discount rate and E_{π_m} represents the expected value of the policy π_m . Action-value function under an ideal policy is signified as $Op_{m, \pi_m^*}(state_{m, r}, action_{m, r})$, expressed in Eq. (7)

$$Op_{m, \pi_m^*}^*(state_{m, r}, action_{m, r}) = \max_{\pi_m} Op_{m, \pi_m}(state_{m, r}, action_{m, r}) \quad (7)$$

In Q-learning, an agent estimates $Op_{m, \pi_m^*}^*$ using a Q-table and chooses an action using one of two methods. In the exploration phase, the agent chooses an action at random with probability ϵ to investigate action space and state space; in the optimization or exploitation phase, the agent chooses an action based on the current state, produces maximum Q-value with probability $1 - \epsilon$. This process is known as greedy action selection²⁷. The greedy action selection mechanism in HEMS focuses on short-term cost savings by choosing actions with maximum short-term rewards, like operating appliances during periods of lowest energy costs. It facilitates fast decision-making and keeps operations simple, making it highly suitable for peak pricing hours. Yet, an exclusive greedy approach can result in inefficient energy allocation since it doesn't always look into long-term energy efficiency or user comfort settings. HEMS, in its practical use, needs to accompany the greedy solution with adaptive learning methods that enhance energy saving to a maximum while preserving user comfort and guaranteeing optimal appliance scheduling over time. Greedy behaviour $action_{m, r}^*$ can be identified by in Eq. (8)

$$action_{m, r}^* = \arg \max_{action} Op_m(state_{m, r}, action) \quad (8)$$

The following update rule is how Q-learning can approach Op_m^* given the action selection strategies.

$$Op_m(state_{m, r}, action_{m, r}) \leftarrow (1 - \alpha) Op_m(state_{m, r}, action_{m, r}) +$$

$$\alpha \left(reward_{m, r+1} + \gamma \max_{action} Op_m(state_{m, r+1}, action) \right) \quad (9)$$

Where, $\alpha \in (0; 1]$ represents learning rate. In the smart home scenario, take $reward$ as a penalty because the user needs to minimize both user dissatisfaction and cost. As a result, the previously stated maximization is substituted with reduction. Home appliance is referred to as agent. m , and its state $state_{m, r}$ is designed in this way.

$$state_{m, r} = [\lambda_r^{RTP}, r] \quad (10)$$

Let $action_{m, r}$ represent the action taken by the agent m in time slot r where λ_r^{RTP} represents the cost of purchasing electricity in real time. Refrigerators have only one action space option, "on", thus they cannot be planned. For washing machine appliances, the action space offers "on" and "off" options to control their operation. The action space for air conditioner appliances consists of discrete numbers. $\{1, 2, 3, \dots, Energy_{m, r}^3\}$. Every element indicates a level of energy consumption, with $Energy_{max}^3$ being maximum level. Agents can obtain two reward signals as a response from their environment^{28,29}.

The rewards for refrigerators (R) are as follows in Eq. (11)

$$reward R1_{m, r} = \lambda_r^{RTP} Energy_{m, r}^1,$$

$$reward R2_{m, r} = U_{m, r}^1 \quad (11)$$

The rewards for washing machine (W) are as follows in Eq. (12).

$$reward W1_{m, r} = \lambda_r^{RTP} Energy_{m, r}^2,$$

$$reward W2_{m, r} = U_{m, r}^2 \quad (12)$$

The reward is applied in the following ways to air conditioner (A) is expressed in Eq. (13)

$$reward A1_{m, r} = \lambda_r^{RTP} Energy_{m, r}^3,$$

$$reward A2_{m, r} = U_{m, r}^3 \quad (13)$$

By changing Q-value $Q_{k,m}(State_{m,r}, action_{m,r})$, the two reward functions outlined above are intended to incentivize agents to minimize objectives expressed in Eq. (14)

$$Q_{k,m}(State_{m,r}, action_{m,r}) \leftarrow (1 - \alpha) Q_{k,m}(State_{m,r}, action_{m,r}) + \alpha \left(h_{k,m,r+1} + \gamma \min_{action} Q_{k,m}(State_{m,r+1}, action) \right) \quad (14)$$

Where, $k \in \{1,2\}$ signifies the k th objective

$$Energy_{m,h} = \sum_{n=1}^3 Energy_{m,r}^n \quad (15)$$

In terms of MORL, the agent m 's reward function in a time slot r is represented as follows: the weighted sum of rewards in (11), (12), and (13).

$$reward_{m,r} = w \cdot \lambda_r^{RTP} Energy_{m,r} + (1 - w) U_{m,r} \quad (16)$$

Where w a weighting coefficient and by using (9), the associated Q-table was updated.

DQN with SAPOA

Smart HEMS, with a combination of DQN and SAPOA, enhances home automation by addressing energy consumption variability and user preferences. This approach promotes efficiency and cost-effectiveness while being convenient for users. SAPOA's dynamic energy management focuses on energy-intensive appliances to reduce consumption and cost. The DQN framework optimizes appliance schedules, improving overall system performance. Performance metrics monitor the efficacy of this design, focusing on flexibility and responsiveness to real-time energy price and user needs changes³⁰. Figure 2 presents the DQN with SAPOA architecture.

Modelling the environment

- Formulation: Consider the HEMS as a place where the Deep Reinforcement Learning (DRL) agent receives the states and incentives. Through action, the agents in a HEMS learn the best behaviours. An agent's goal is to maximize discounted cumulative reward by learning a policy (s) over a collection of states to a probability distribution. Mathematical description is expressed in Eq. (17).

$$Reward_{ft} = \sum_{t=0}^{T_m} \gamma^t reward(State_t, act_t) \quad (17)$$

where ft is a future reward in the interval $[0,1]$ and $Reward_{ft}$ is a reward function. DQN's goal is to ascertain the best course of action for maximizing $Reward_{ft}$. When action act_{tm} is in state $State_{tm}$, using policy, and Q-value $Q(State_t, act_t)$, reward $Reward_{ft}$ is defined. Quality of a certain activity in a specified condition is represented by the Q-value. a fully associated DNN with a weight set that is used to approximate Q-function that the DQN model uses. Agent receives the reward $Reward_{ft}$ for each activity, after which the HEMS is moved to the next stage $State_{tm+1}$ ³¹. For the state $State_{tm}$, action with maximum Q value is chosen.

$$act_{tm} = \operatorname{argmax} Q(State_{tm}, act_{tm}) \quad (18)$$

Equation (18), where $[0,1]$ is the learning rate and Q' denotes the improved policy, and describes the enhanced policy with Q-values. The DQN model's input state data at time tm in state space State is $State_{tm}$, where S stands for an array value and a series of the network's adjacency matrix, data size, and cluster-ID. Mini-batches $(s, a, r, s') \sim U(D)$ randomly selected from the memory pool can be used to train DNN, which helps to

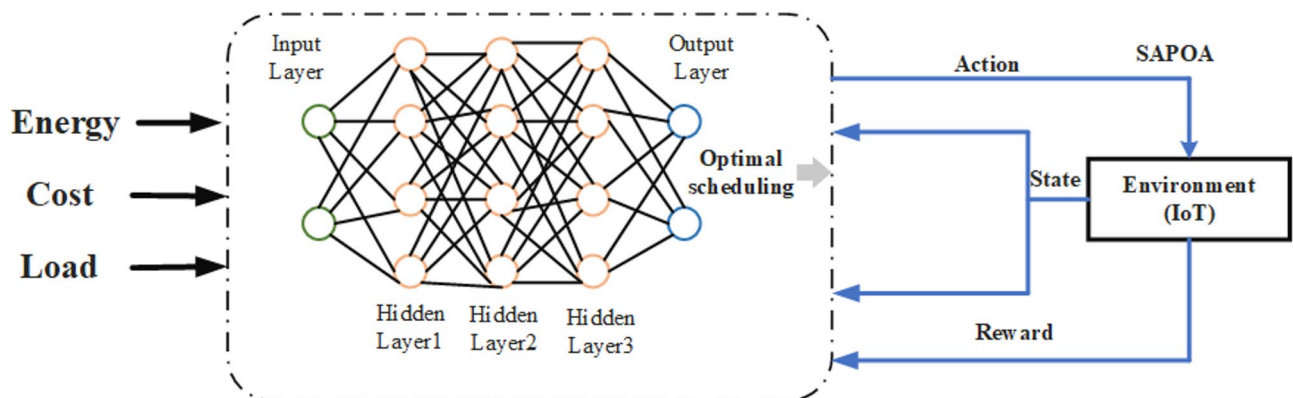


Fig. 2. DQN with SAPOA architecture.

ensure system stability and remove strong correlations between data. The Bellman equation, which is used to train the target value for every state-action combination in Q learning, serves as the loss function of DNNs and is expressed as an Eq.

$$y_t = r(\text{State}_{tm}, \text{act}_{tm}) + \gamma Q(\text{State}_{tm+1}, \pi(\text{State}_{tm+1}) | \theta^Q) \quad (19)$$

where the DQN parameter is θ^Q . The DQN is trained with the target value, which can be achieved through loss function minimization. It can be stated as follows:

$$L(\theta^Q) = E[y_t - Q(\text{State}_{tm}, \text{act}_{tm} | \theta^Q)] \quad (20)$$

DQN algorithm struggles due to persistent problems and state space growth. Actor-critic RL method is applied to address this issue⁵². Actor and critic functions are utilized to enhance algorithm learning. Critic evaluates actions, while the actor selects actions. Updating the actor-network involves calculating cumulative reward with the chain rule function in Eq. (21).

$$\nabla_{\theta^p} \approx e[\nabla_{\theta^p} Q(s, a | \theta^Q | s = \text{State}_{tm}, a = \pi(\text{State}_{tm} | \theta^p))] \quad (21)$$

- States: HEMS states represent the current system status, enabling the agent to make informed decisions for future reward optimization. The agent controls device switching or adjustment based on existing home automation systems [28–30].
- Actions: The Q-value combining action, determined by the neural network, manages energy use aspects by:
 - Controlling power consumption and shifting loads.
 - Adjusting on/off times for loads.
 - Regulating charging/discharging processes of energy storage systems This strategic control balances load storage, optimizes energy management dynamically, enhances cost efficiency and user satisfaction, and adapts to changing energy consumption patterns and user preferences.
- Rewards: The Self-Adaptive Puma Optimizer Algorithm (SAPOA) in the augmented RL enables optimal decision-making based on environmental data, maximizing cost savings and user satisfaction while adapting to dynamic changes in energy use, costs, and user preferences.
- Optimization objectives and parameters: Reduce energy consumption, costs, and user satisfaction by optimizing appliance scheduling and lowering the PAR to improve grid stability and minimize disruptions. The optimization process involves adjusting static parameters like α , K , and β to balance exploration and exploitation, while P_1^r and P_2^r guide fast-running or ambush strategies, and measure energy consumption from appliances.

Experimentation

An agent of a smart home is HEMS. The State is configured as follows in Eq. (22).

$$\text{state}_r^{ess} = [B_r, \lambda_r^{RTP}, r] \quad (22)$$

The battery system agent's activity, represented by a_r^{ess} , is P_r . This agent receives feedback from the surroundings as its only reward signal, in contrast to the agents connected to the household appliances.

$$\text{reward}_r^{ess} = \lambda_r^{RTP} P_r \Delta r \quad (23)$$

Since the agent wants to lower the energy bill, the reward is the cost of the electricity. The following expression updates the agent's Q-network.

$$Q^{ess}(\text{state}_r^{ess}, \text{act}_r^{ess}) \leftarrow (1 - \alpha) Q^{ess}(\text{state}_r^{ess}, \text{act}_r^{ess}) + \alpha \left(\text{reward}_{r+1}^{ess} + \gamma \min_a Q^{ess}(\text{state}_r^{ess}, \text{act}_r^{ess}) \right) \quad (24)$$

For the best parameter selection, the SAPOA is presented with DQN, which facilitates fine-tuning to produce the most appropriate results.

Integration with SAPOA

SAPOA is an advanced algorithm, based on puma optimization algorithm (POA) integrated with reinforcement learning techniques, specifically designed for smart HEMS. The Puma, also known as a cougar or mountain lion, is the largest subtle cat species in the Americas, inhabiting the Andes to the Yukon. They are adaptable predators with large territories, but human activities have endangered their populations. Pumas are agile, weigh 105.2 kg, and hunt using their powerful leap and fangs. Males mark territories that overlap with females, and their territories depend on prey abundance and vegetation²¹.

- Mathematical model: A mathematical model for a puma optimization algorithm is presented, inspired by puma behaviours in nature. The optimizer algorithm changes between exploration and exploitation phases,

mimicking puma behaviour. Different mechanisms are used for optimization in each phase, similar to how pumas live in nature. Mathematical formulations are provided to explain the optimization operations in the algorithm.

- Puma intelligence (phase change mechanism): Pumas, intelligent and memory-rich, hunt in familiar areas based on past experiences. A new hyper-heuristic algorithm, inspired by pumas' intelligence and memory, uses diversity and intensification components. The phase change mechanism is inspired by cougars' hunting strategies, where less experienced cougars explore new areas and ambush promising ones³³.
- New phase: Puma lacks experience early on and explores its territory while hunting in favorable areas. The Puma algorithm combines exploration and exploitation initially, using only two functions in the first three iterations. The process involves calculating fn_1 and fn_2 with specific equations.

$$fn_{1exp} = a \cdot \left(\frac{S_{Cexp}^1}{S_t} \right) \quad (25)$$

$$fn_{1exl} = a \cdot \left(\frac{S_{Cexl}^1}{S_t} \right) \quad (26)$$

$$fn_{2exp} = b \cdot \left(\frac{S_{Cexp}^1 + S_{Cexp}^2 + S_{Cexp}^3}{S_{t1} + S_{t2} + S_{t3}} \right) \quad (27)$$

$$fn_{2exl} = b \cdot \left(\frac{S_{Cexl}^1 + S_{Cexl}^2 + S_{Cexl}^3}{S_{t1} + S_{t2} + S_{t3}} \right) \quad (28)$$

Equations (25)–(37) are used to determine the variable values of SC , which are associated with each stage of exploration and exploitation. Additionally, S_t is a variable that has a constant value, which is 1 in our case. The fixed-value parameters a and b , which are used to rank each of the fn_1 and fn_2 functions must be set before the optimization process.

$$S_{Cexp}^1 = |C_{best}^{Ini} - C_{exp}^1| \quad (29)$$

$$S_{Cexp}^2 = |C_{exp}^2 - C_{exp}^1| \quad (30)$$

$$S_{Cexp}^3 = |C_{exp}^3 - C_{exp}^2| \quad (31)$$

$$S_{Cexl}^1 = |C_{best}^{Ini} - C_{exl}^1| \quad (32)$$

$$S_{Cexl}^2 = |C_{exl}^2 - C_{exl}^1| \quad (33)$$

$$S_{Cexl}^3 = |C_{exl}^3 - C_{exl}^2| \quad (34)$$

In Eqs. (38)–(43), C_{best}^{Ini} represents the cost of the top solution in the initialization phase with six variables. $C_{exp}^1, C_{exp}^2, C_{exp}^3, C_{exl}^1, C_{exl}^2, C_{exl}^3$ for both exploration and exploitation phases are calculated. The functions fn_1 and fn_2 are evaluated after the 3rd iteration, leading to the selection of either the exploration or exploitation phase³⁴. Pumas make a choice based on pleasant experiences to determine which phase to proceed with.

Equations (25)–(37) and Eqs. (38)–(43) help balance exploration and exploitation in an algorithm, resembling pumas' hunting behavior. They assess solution performance during exploration and exploitation, deciding whether to focus on exploration or shift to exploitation based on past and current outcomes.

The points for exploitation and exploration are determined using Eqs. (35) and (36) respectively.

$$SC_{exp} = (a \cdot fn_{1exp}) + (b \cdot fn_{2exp}) \quad (35)$$

$$SC_{exl} = (a \cdot fn_{1exl}) + (b \cdot fn_{2exl}) \quad (36)$$

SC_{exp} and SC_{exl} , calculated with Eqs. (35) and (36), determine exploration and exploitation phases. If $SC_{exl} > SC_{exp}$, exploitation stage is entered; otherwise, the exploration step is taken. Each step independently finds solutions after 3 iterations, exceeding the total population. The total cost of solutions from both phases is calculated at the end of 3rd iteration to address the issue. Only the best solutions from all produced solutions replace current solutions, equalling the entire population.

Experimentation phase

After three generations, Pumas decides to change the phase and chooses one phase for optimization. Three functions (fn_1, fn_2, fn_3) are used for scoring. The first function emphasizes the escalation component, prioritizing exploration or exploitation phases. The exploration phase is more emphasized in the first function, calculated using Eq. (37).

$$fn_{1itr}^{exl} = a \cdot \left| \frac{C_{past}^{exl} - C_{recent}^{exl}}{Itr_{itr}^{exl}} \right| \quad (37)$$

$$fn_{1itr}^{exp} = a \cdot \left| \frac{C_{past}^{exp} - C_{recent}^{exp}}{Itr_{itr}^{exp}} \right| \quad (38)$$

In Eqs. (37) and (38), fn_{1itr}^{exl} and fn_{1itr}^{exp} represent the amount of the first function based on exploration or exploitation, with itr as the iteration number. Costs depict the best solution before and after improvements in the current choice. Itr_{itr}^{exp} and Itr_{itr}^{exl} are the unselected iterations from the previous to the current selection. a is a user-adjustable parameter prioritizing the first function based on its value.

$$fn_{2itr}^{exl} = b \cdot \left| \frac{(C_{past,1}^{exl} - C_{recent,1}^{exl}) + (C_{past,2}^{exl} - C_{recent,2}^{exl}) + (C_{past,3}^{exl} - C_{recent,3}^{exl})}{Itr_{itr,1}^{exl} + Itr_{itr,2}^{exl} + Itr_{itr,3}^{exl}} \right| \quad (39)$$

$$fn_{2itr}^{exp} = b \cdot \left| \frac{(C_{past,1}^{exp} - C_{recent,1}^{exp}) + (C_{past,2}^{exp} - C_{recent,2}^{exp}) + (C_{past,3}^{exp} - C_{recent,3}^{exp})}{Itr_{itr,1}^{exp} + Itr_{itr,2}^{exp} + Itr_{itr,3}^{exp}} \right| \quad (40)$$

The second function emphasizes resonance and performance, aiding the exploitation phase. It is calculated using Eqs. (39) and (40). fn_{2itr}^{exl} and fn_{2itr}^{exp} represent the second function in exploration or exploitation, with t as the iteration number. Costs are related to the best solutions before and after improvements in selections. Itr_{itr}^{exl} and Itr_{itr}^{exp} are the unselected iterations in each phase. b is a parameter determining the adequacy of the second function.

$$fn_{3itr}^{exl} = \begin{cases} \text{if selected, } fn_{3itr}^{exl} = 0 \\ \text{otherwise, } fn_{3itr}^{exl} + c \end{cases} \quad (41)$$

$$fn_{3itr}^{exp} = \begin{cases} \text{if selected, } fn_{3itr}^{exp} = 0 \\ \text{otherwise, } fn_{3itr}^{exp} + c \end{cases} \quad (42)$$

The third function in the selection mechanism highlights diversity and gives a chance for unselected phases to be chosen. It is defined in Eqs. (41) and (42). The function relates to exploitation or exploration stages based on the iteration number. The value of the function increases by parameter c if a stage is not chosen. c must be set between 0 and 1. A higher c value increases the chances for low-score phases.

$$Fn_{itr}^{exl} = (\alpha_{itr}^{exl} \cdot (fn_{1itr}^{exl})) + (\alpha_{itr}^{exl} \cdot (fn_{2itr}^{exl})) + (\delta_{itr}^{exl} \cdot (Calc \cdot fn_{3itr}^{exl})) \quad (43)$$

$$Fn_{itr}^{exp} = (\alpha_{itr}^{exp} \cdot (fn_{1itr}^{exp})) + (\alpha_{itr}^{exp} \cdot (fn_{2itr}^{exp})) + (\delta_{itr}^{exp} \cdot (Calc \cdot fn_{3itr}^{exp})) \quad (44)$$

$$Calc = \{ \{|C_{past} - C_{recent}| \}^{exl}, \{|C_{past} - C_{recent}| \}^{exp} \}, 0 \notin Calc \quad (45)$$

$$\alpha_{itr}^{exp,exl} = \begin{cases} \text{if } Fn_{itr}^{exl} > Fn_{itr}^{exp}, \alpha^{exl} = 0.99, \alpha^{exp} = \lfloor \alpha^{exp} - 0.01, 0.01 \rfloor \\ \text{otherwise, } \alpha^{exp} = 0.99, \alpha^{exl} = \lfloor \alpha^{exl} - 0.01, 0.01 \rfloor \end{cases} \quad (46)$$

$$\delta_{itr}^{exl} = 1 - \alpha_{itr}^{exl} \quad (47)$$

$$\delta_{itr}^{exp} = 1 - \alpha_{itr}^{exp} \quad (48)$$

The cost of phase change is calculated using Eqs. (47) and (48). Final costs for each phase are computed. Parameters α and δ change during the search operation.

$$\text{If } r_1 > 0.5, G_{soln} = r_d * (u - l) + l, \text{ Otherwise} \quad (49)$$

$$G_{soln} = P_{1, soln} + Soln \cdot (P_{1, soln} - P_{2, soln}) + Soln \cdot (((P_{1, soln} - P_{2, soln}) - (P_{3, soln} - P_{4, soln})) + ((P_{3, soln} - P_{4, soln}) - (P_{5, soln} - P_{6, soln}))) \quad (50)$$

$$Soln = 2 \cdot r_2 - 1 \quad (41)$$

If the exploration cost exceeds the exploitation cost, a parameter of α exploitation is penalized. If the exploitation cost is higher, the procedure is reversed. $Calc$ represents cost differences between exploitation and exploration improvements.

$$P_{recent} = \begin{cases} G_{soln}, \text{ if } j = j_r \text{ or } r_3 < Opt \\ P_{1, soln}, \text{ otherwise} \end{cases} \quad (52)$$

$$nc = 1 - Opt \quad (53)$$

$$H = \frac{nc}{n_p} \quad (54)$$

$$CP_{recent} < CP_{i, Opt} = Opt + H \quad (55)$$

Using pumas' behavior in finding food, new solutions are generated. The search agents are sorted by cost, with high-quality solutions prioritized initially. As the parameter increases, solutions with higher cost values change to explore worse solutions. Good-quality solutions undergo a few changes to avoid local optimality traps while ensuring the new production Pumas is better than the existing ones³⁵.

$$P_{1, Soln} = P_{recent}, \text{ if } P_{i, recent} < P_{1, Soln} \quad (56)$$

Newly generated solutions replace current ones if they offer better cost.

Exploitation phase

In the exploitation stage, the PO algorithm uses two operators to enhance solutions based on pumas' hunting behaviours. Pumas either ambush prey or sprint after them in nature. When hunting in the first mode, pumas use case 1 in Eq. (48) for running and ambush strategies. If r_4 is greater than 0.5, the fast-running strategy is used; otherwise, the ambush strategy is selected. The ambush strategy involves short and long jumps towards prey.

if $r_4 > 0.5$,

$$P_{recent} = \frac{\left(\frac{\text{mean}(A_{soln})}{n_p} \right) \cdot P_1^r - (-1)^\beta \times P_i}{1 + (\alpha \cdot r_5)}$$

if $r_6 > K$,

$$P_{recent} = Puma_M + (2 \cdot r_7) \cdot Ex(rN_1) \cdot P_2^r - P_i$$

Or in general,

$$P_{new} = (2 \times r_8) \times \frac{(Fn_1 \cdot r \cdot P(i) + Fn_2 \cdot (1-r) \cdot Puma_M)}{(2 \cdot r_9 - 1 + rN_2)} - Puma_M$$

(48)

Parameters α , K , and β are static and must be adjusted before optimization. Puma male is the best solution for the population, and various random numbers are generated during the process. Additionally, Ex , rN_1 , rN_2 , and P_2^r play roles in the optimization procedure. The SAPOA optimizes energy scheduling dynamically by varying decision variables according to real-time situations. This helps in effective appliance scheduling with the least cost. Integration with Bernoulli distribution Eq. (57) helps in adaptability so that the system can efficiently react to user needs and dynamic pricing, reducing wastage of energy and grid stability.

$$B_{fn}(P_i, p) = p^{P_i} (1-p)^{1-P_i} \text{ for } P_i \in \{0,1\} \quad (57)$$

Thus by using Bernoulli distribution Eqs. (57) in Eqs. (56),

if $r_4 > 0.5$,

$$P_{recent} = \frac{\left(\frac{\text{mean}(A_{soln})}{n_p} \right) \cdot P_1^r - (-1)^\beta \times B_{fn}(P_i, p)}{1 + (\alpha \cdot r_5)}$$

if $r_6 > K$,

$$P_{recent} = Puma_M + (2 \cdot r_7) \cdot Ex(rN_1) \cdot P_2^r - B_{fn}(P_i, p)$$

The general form is as,

$$P_{new} = (2 \times r_8) \times \frac{(Fn_1 \cdot r \cdot B_{fn}(P_i, p) + Fn_2 \cdot (1-r) \cdot Puma_M)}{(2 \cdot r_9 - 1 + rN_2)} - Puma_M$$

(50)

Combining the Puma Optimizer Algorithm with a Bernoulli distribution enhances dynamic control over parameters and optimization efficiency through POA-based self-adaptive optimization, contributing to an improvement in overall performance³⁶.

Flow chart for DQN with SAPOA

Figure 3 demonstrates the flowchart for DQN with SAPOA.

Result and discussion

This section includes results and an outline of the proposed model. This examination gives an innovative DR method that is presented using a multi-objective enhanced reinforcement learning model called DQN with a SAPOA. The version's performance was assessed using measures together with PAR, Electricity cost, and Energy consumption.

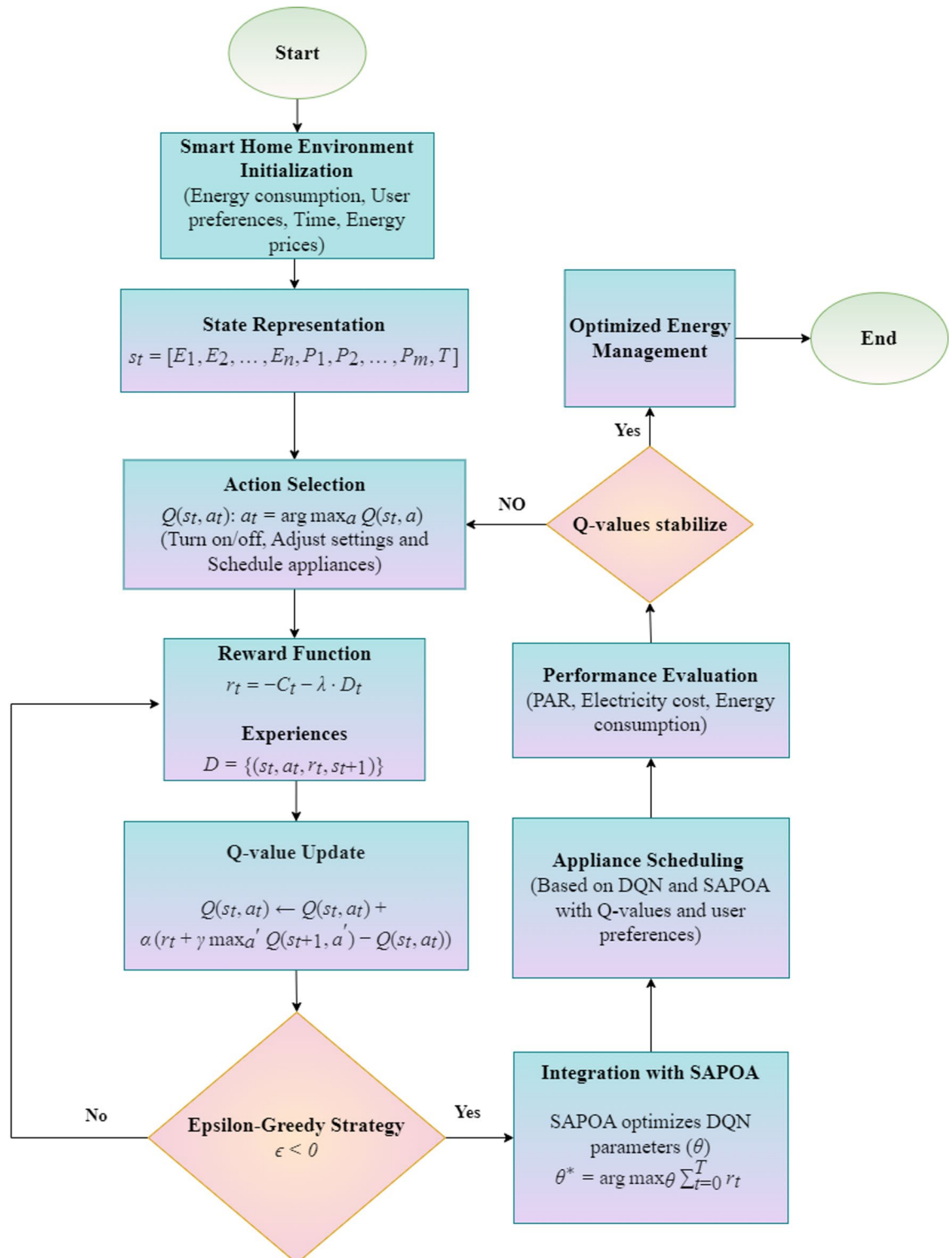


Fig. 3. Flowchart for the DQN with SAPOA.

Experimental and simulation setup

The machine has an Intel Core i5GHz, 12 GB of RAM, and a 64-bit version of Windows 10. Installed is MATLAB 2022b, which offers a stable environment for algorithmic and computational work. This configuration guarantees sufficient processing power and memory to efficiently conduct complicated simulations and analyses. Table 2 presents simulation parameters and details over MATLAB environment²⁷.

The HEMS simulation environment mimics real conditions with realistic patterns of energy consumption, dynamic pricing schemes, and renewable sources of energy. It replicates typical user behavior, has feedback mechanisms, and uses performance metrics like total active time PAR to gauge efficacy. It uses past energy consumption data to model typical appliance usage and models energy price fluctuations as a function of time-of-use pricing. Renewable energy sources, such as solar production, are also viewed to offset their volatility. MORL maximizes DR by finding a balance between energy expenses, user comfort, and grid stability. MORL’s dual Q-table method learns to schedule appliances dynamically with real-time energy prices and user satisfaction, efficiently adapting to dynamic energy situations and overall energy efficiency and user satisfaction.

Performance metrics

Several matrices, including PAR, Electricity cost, and Energy consumption are used for performance assessment.

- PAR: It is frequently used in signal processing to calculate the difference between a signal’s peak and average values. It is used by digital communication systems to calculate signal strength and dynamic range.

$$PAR = \frac{Peak\ value}{Average\ Value} \tag{58}$$

- Energy consumption: Depending on the circumstance and the energy source, different methods are employed to calculate energy consumption.

$$Energy\ (E) = Power\ (P) \times Time\ (t) \tag{59}$$

- Electricity cost: It is the cost of purchasing electricity for personal use. The factors that influence electricity prices are location, energy sources, market conditions, regulatory policies, and consumption ways.

$$Cost = Energy \times Rate \tag{60}$$

Performance analysis

Table 3 describes the PAR values for various energy management scenarios with and without RES. An unscheduled scenario showed a high value of PAR to be 3.4286, showing the inefficient usage of energy. The proposed method reduces the PAR value to 1.9765 in the absence of RES and further downtick to 1.0339 with RES, hence more efficient. Further, the values reached by MORL-POA and SAPOA in the absence of RES were 2.1877 and 2.2789 respectively, while POA, remained at the base value. While MORL-POA and SAPOA increased to 1.0882 and 1.1195 off RES, they were still behind the suggested approach. An increase of 1.2155 was measured by POA using RES.

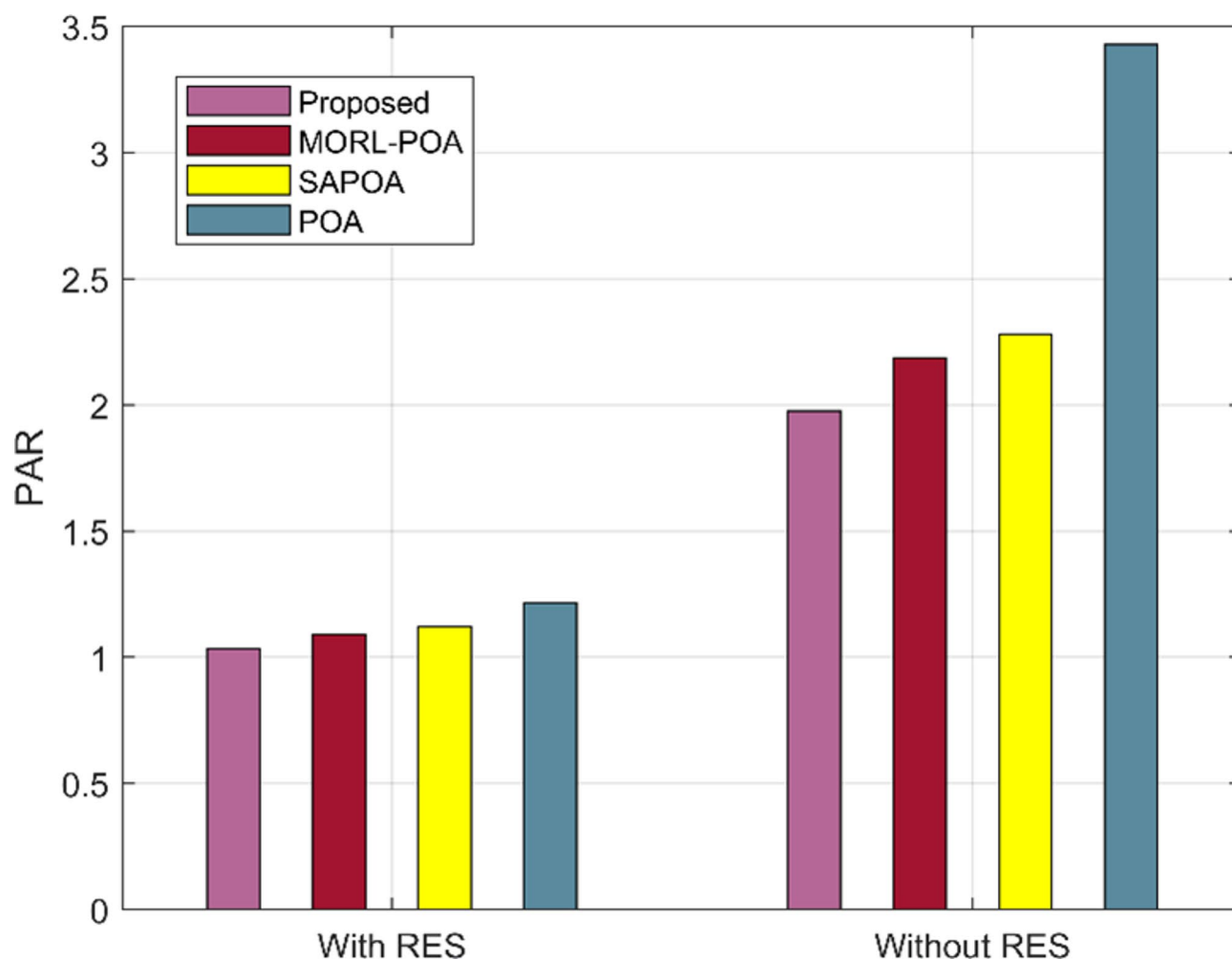
Figure 4 shows the Graphical representation of the PAR comparison with unscheduled, Proposed without RES and with RES.

Table 4 compares the different scenarios of energy management with and without RES in terms of electricity costs. The unscheduled scenario leads to a high cost of 92.915 units. If the proposed technique is used without RES, then this cost comes down drastically to 26.47 units. Further introduction of RES reduces this cost drastically to 6.4179 units, which is highly cost-effective. Even without RES, the proposed method remains the

Component	Parameter/Setting	Description
MATLAB Version	R2021b or later	For RL, optimization, and deep learning support.
Toolboxes	RL, Optimization, Deep Learning, Simulink (optional)	Required for RL algorithms, optimization, and modelling.
State Space	state = [appliance_consumption, price, battery_state]	Defines energy consumption, price, and battery status.
Action Space	actions = [on, off, shift, charge]	Actions for appliances, energy shifting, and charging.
Reward Function	reward = -cost + satisfaction - penalty	Balances cost minimization, user satisfaction, and penalties.
Q-Network	DQN: input → hidden → output	Neural network structure for Q-value approximation.
RL Agent	agent = rlDQNAgent(qNetwork)	Defines the RL agent using DQN.
Training Params	MaxEpisodes = 1000, DiscountFactor = 0.99, EpsilonDecay = 0.995	Training configuration for agent’s learning.
Optimization Method	SAPOA (Self-Adaptive Puma Optimizer Algorithm)	Optimization for appliance scheduling and energy management.
Optimization Variables	optimvar(‘schedule’, ‘LowerBound’, 0, ‘UpperBound’, 1)	Decision variables for appliance scheduling.
Objective	objective = minimize_cost(x)	Objective function for energy cost minimization.
Platform	MATLAB/Simulink	MATLAB for coding; Simulink for dynamic system simulation.
Data Utilization	Historical/synthetic data	Use energy consumption profiles for training and testing.
Performance Metrics	PAR, energy usage, cost savings	Metrics for evaluating optimization success.

Table 2. Simulation parameters and details over MATLAB environment.

PAR comparison	Unscheduled	3.4286
	Proposed without RES	1.9765
	Proposed with RES	1.0339
Without RES	Proposed	1.9765
	MORL-POA	2.1877
	SAPOA	2.2789
	POA	3.4286
With RES	Proposed	1.0339
	MORL-POA	1.0882
	SAPOA	1.1195
	POA	1.2155

Table 3. PAR comparison and without-with RES.**Fig. 4.** Graphical representation of the PAR comparison with unscheduled, Proposed without RES and with RES.

cheapest, at 26.47 units against 26.625 for MORL-POA, 26.79 for SAPOA, and 27.955 for POA. With RES, it is again very outstanding, with MORL-POA to cost about 6.4481, SAPOA 10.1996, and POA 11.054 units. Graphical representation of the Electricity cost comparison with unscheduled, proposed without RES and with RES is shown in Fig. 5.

Table 5 presents a comparison of energy usage using different approaches with and without the use of Renewable Energy Sources (RES). With an energy consumption of 111.9 kWh, the Unscheduled approach is the least efficient. With a consumption of 34.3 kWh, Proposed is the least energy-intensive planned method without RES. It is followed in order by MORL-POA, which uses 38.3 kWh, SAPOA, and POA, which use 40.8 kWh

Electricity cost comparison	Unscheduled	92.915
	Proposed without RES	26.47
	Proposed with RES	6.4179
Without RES	Proposed	26.47
	MORL-POA	26.625
	SAPOA	26.79
	POA	27.955
With RES	Proposed	6.4179
	MORL-POA	6.4481
	SAPOA	10.1996
	POA	11.054

Table 4. Electricity cost comparison with and without RES.

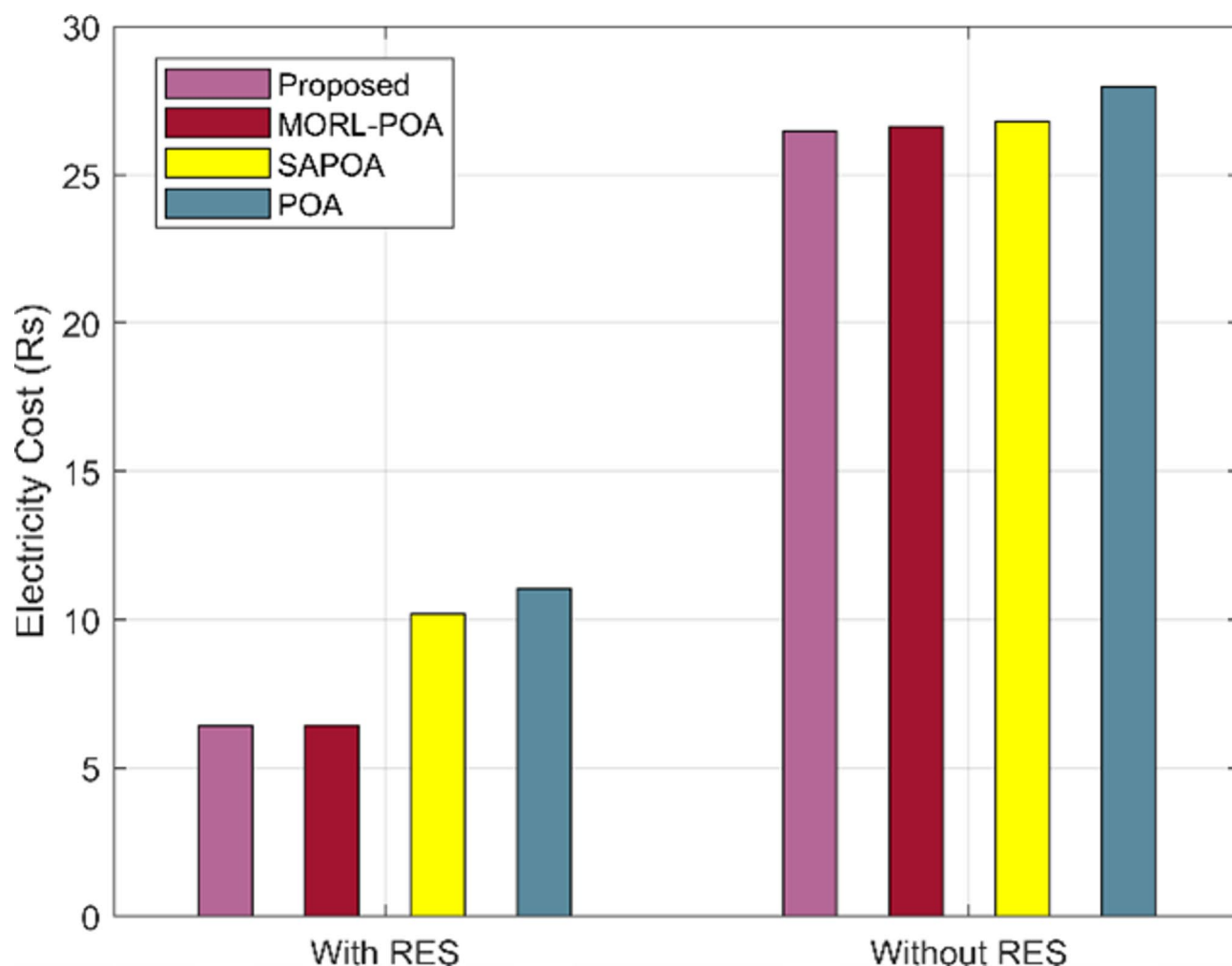


Fig. 5. Graphical representation of the electricity cost comparison with unscheduled, Proposed without RES and with RES.

and 46.4 kWh, respectively. With RES, the Proposed technique continues to be the most efficient, consuming 33.7 kWh, which is marginally less than MORL-POA's 34 kWh, while SAPOA and POA both use 35.7 kWh. According to the statistics, all techniques see a decrease in energy usage when RES is integrated, but the proposed approach has the greatest advantage from this integration. Graphical representation of the Energy Consumption comparison with unscheduled, proposed without RES and with RES is demonstrated in Fig. 6.

The appliances shown in Figs. 7, 8, 9, 10 and 11 such as the washing machine, AC1, AC2, AC3, and refrigerator illustrate varied patterns of energy usage. Regarding the Washing Machine, the Proposed line fluctuates between

Energy consumption comparison	Unscheduled	111.9
	Proposed without RES	34.3
	Proposed with RES	33.7
Without RES	Proposed	34.3
	MORL-POA	38.3
	SAPOA	40.8
	POA	46.4
With RES	Proposed	33.7
	MORL-POA	34
	SAPOA	35.7
	POA	35.7

Table 5. Energy consumption comparison without and with RES.

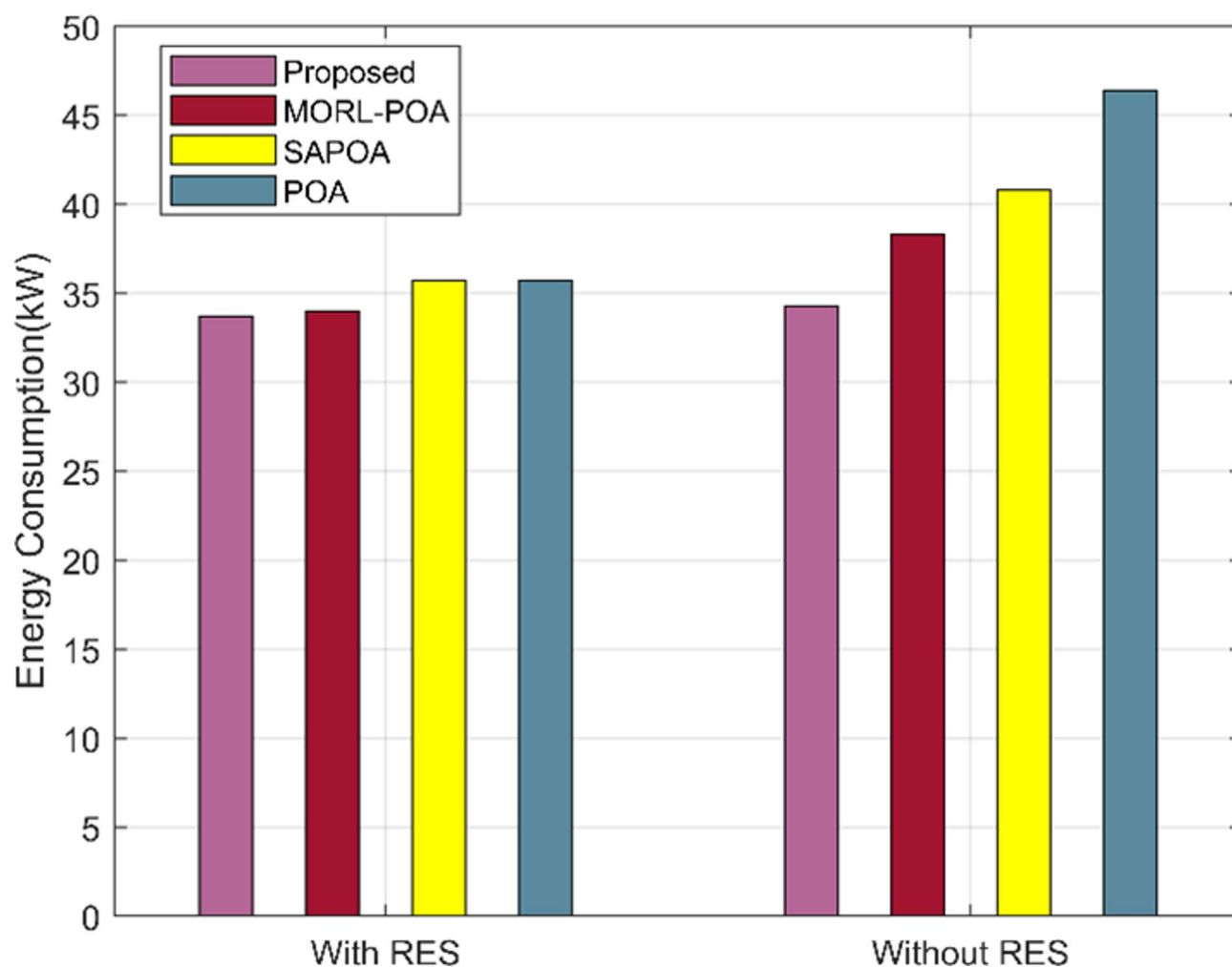


Fig. 6. Graphical representation of the energy consumption comparison with unscheduled, Proposed without RES and with RES.

0.1 and 0.8 units, reaching a peak of approximately 10–12 h, while the Unscheduled line consistently displays a high consumption of about 0.9 units throughout the day, with a minor evening peak.

In AC1, the Proposed line fluctuates between 0.1 and 0.7 units, but the Unscheduled line stays constant at roughly 0.85 units, increasing slightly in the evening. Similar to that, AC2 displays consistent Unscheduled consumption of approximately 0.8 units, but the Proposed line decreases to 0.1 units at peak periods. AC3 exhibits a low Proposed level of 0.1 units in comparison to a consistently high Unscheduled level of approximately 0.9 units. Last but not least, the Refrigerator chart shows significant Unscheduled consumption of approximately 0.9

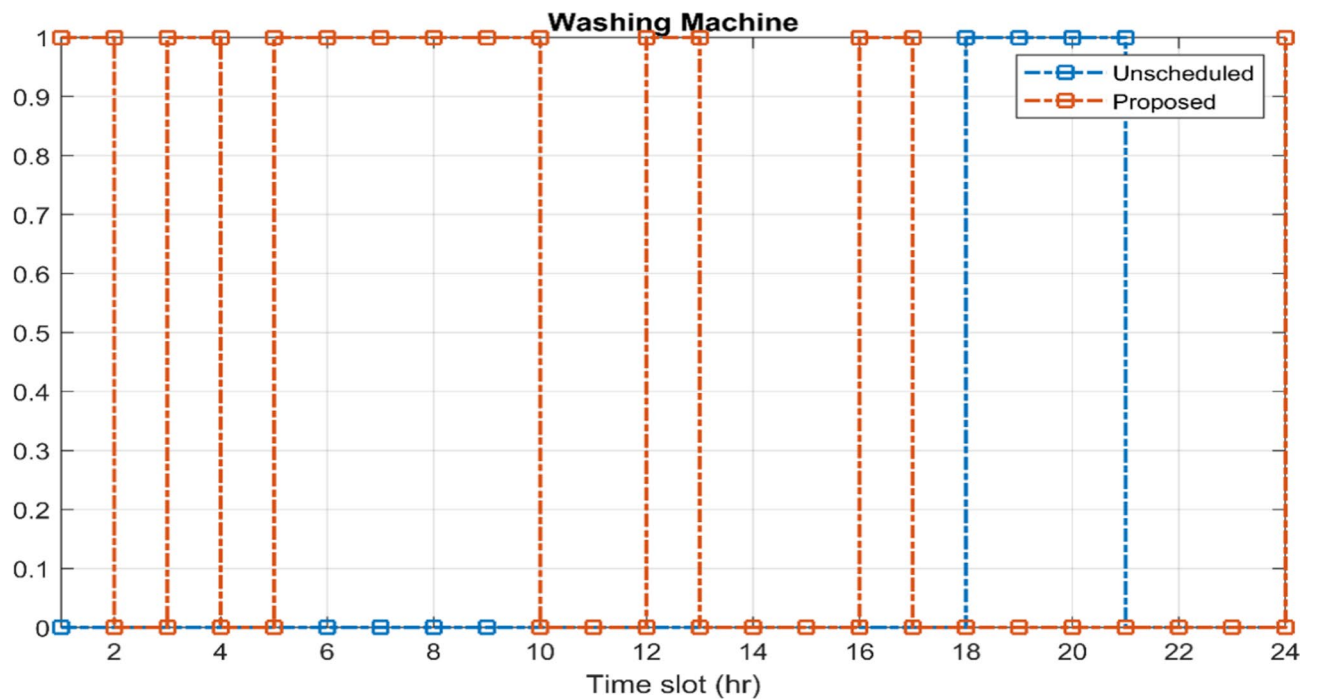


Fig. 7. Comparison of unscheduled and scheduled patterns for washing machine.

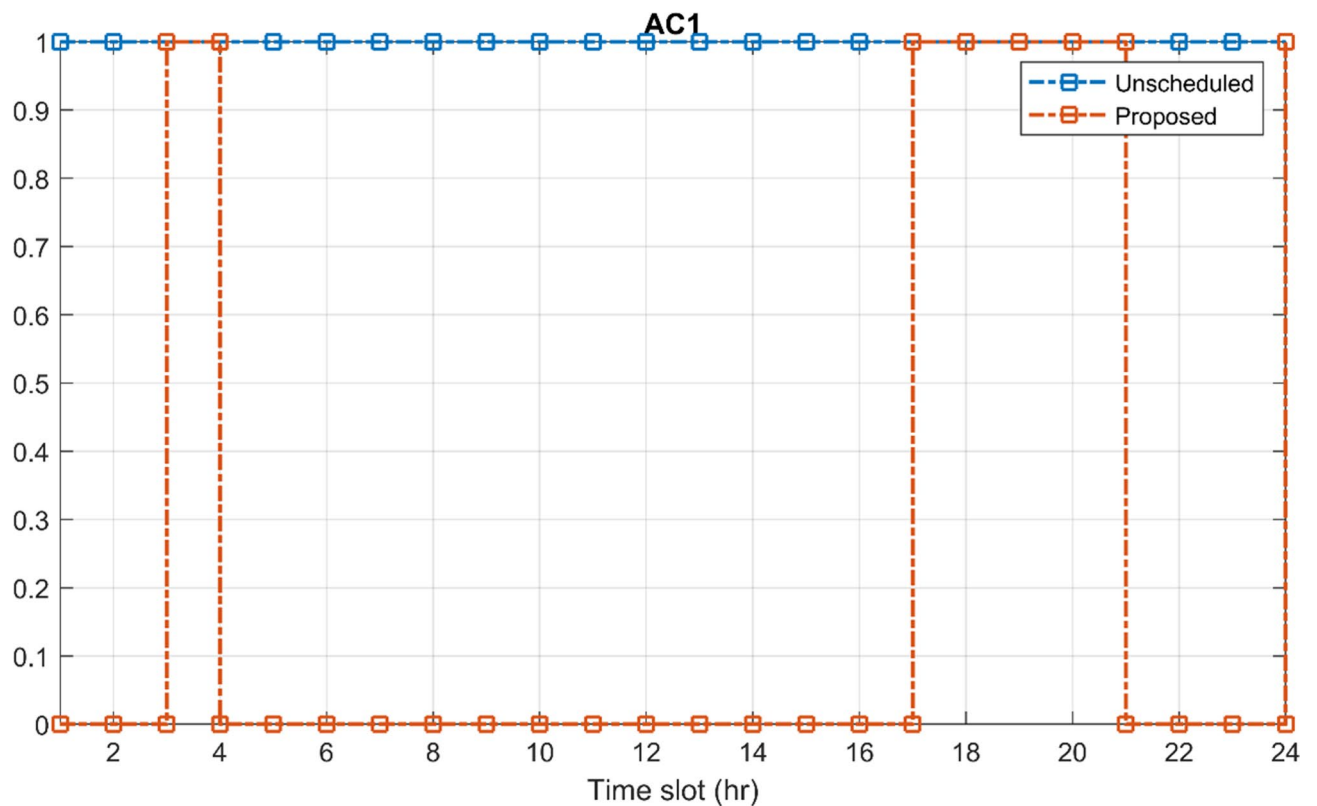


Fig. 8. Comparison of unscheduled and scheduled patterns for air conditioner 1.

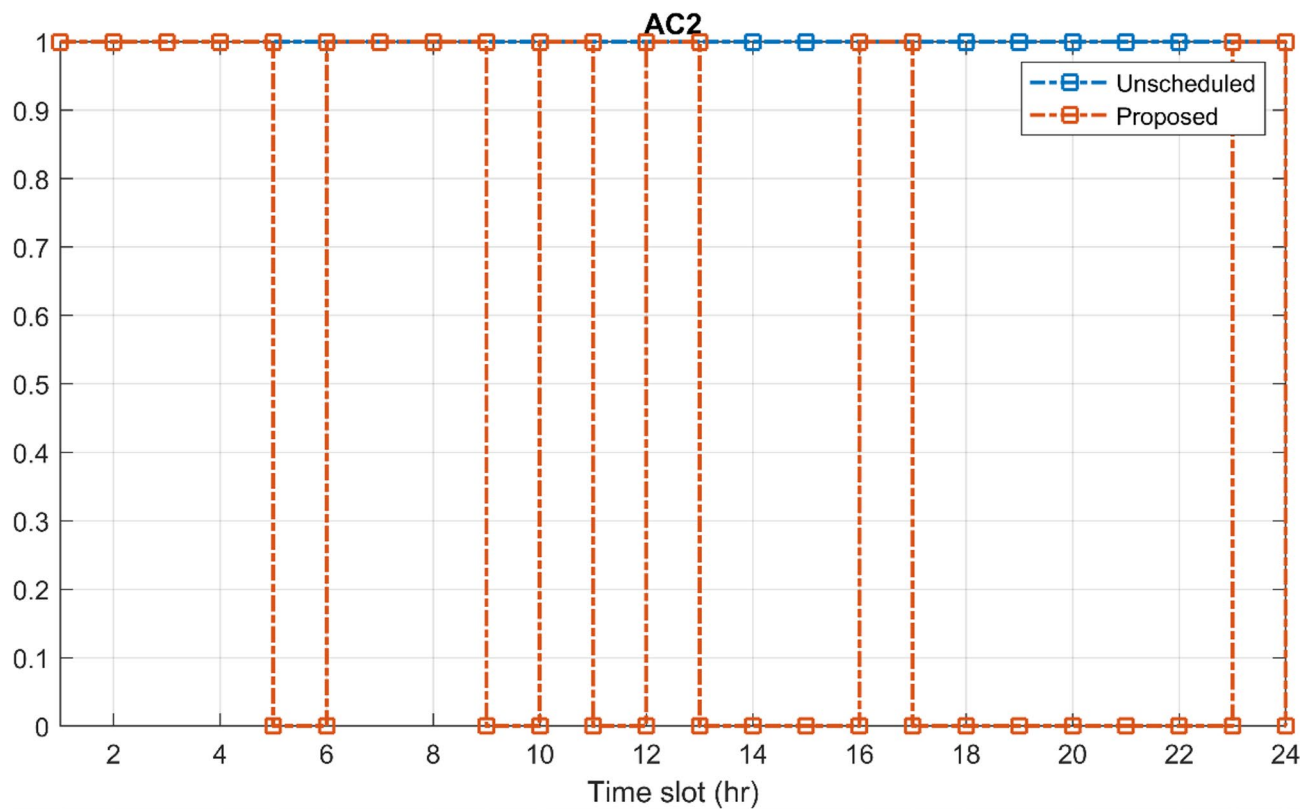


Fig. 9. Comparison of Unscheduled and scheduled patterns for air conditioner 2.

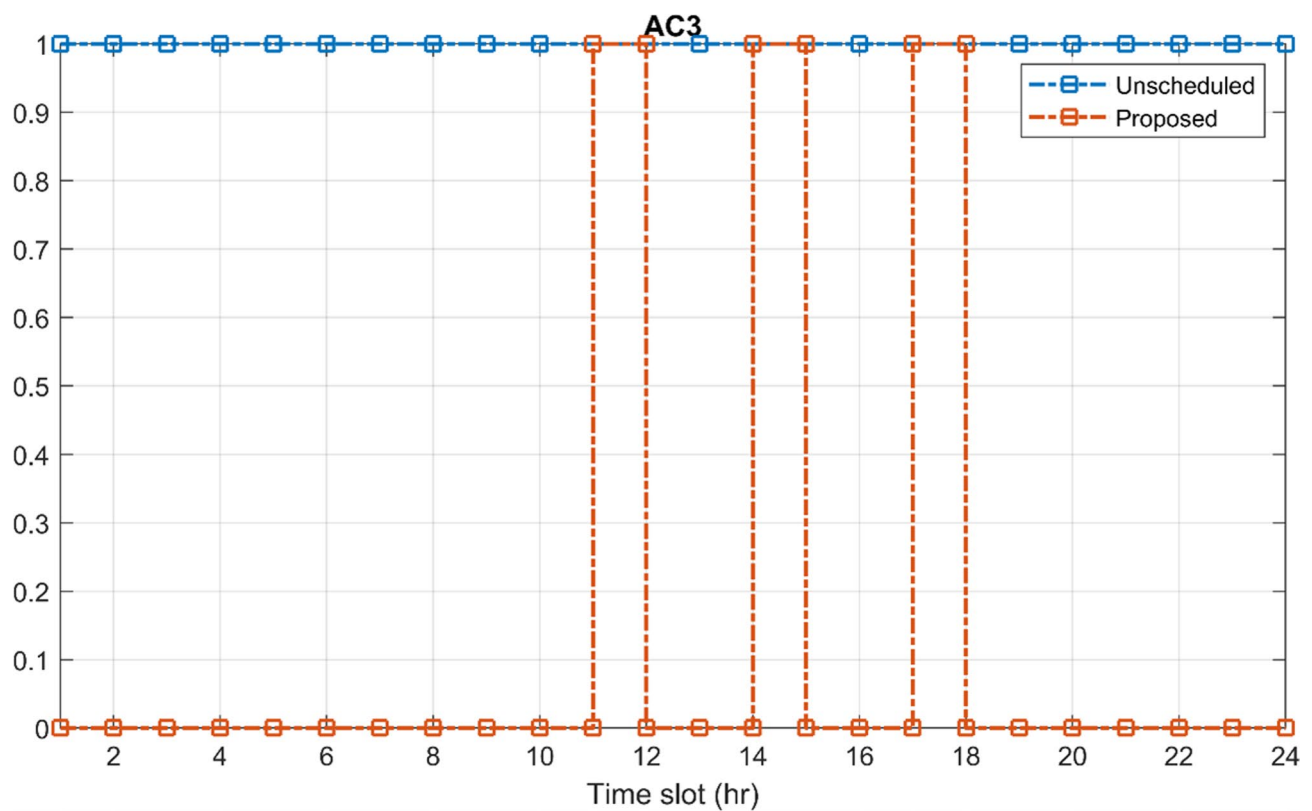


Fig. 10. Comparison of Unscheduled and scheduled patterns for Air conditioner 3.

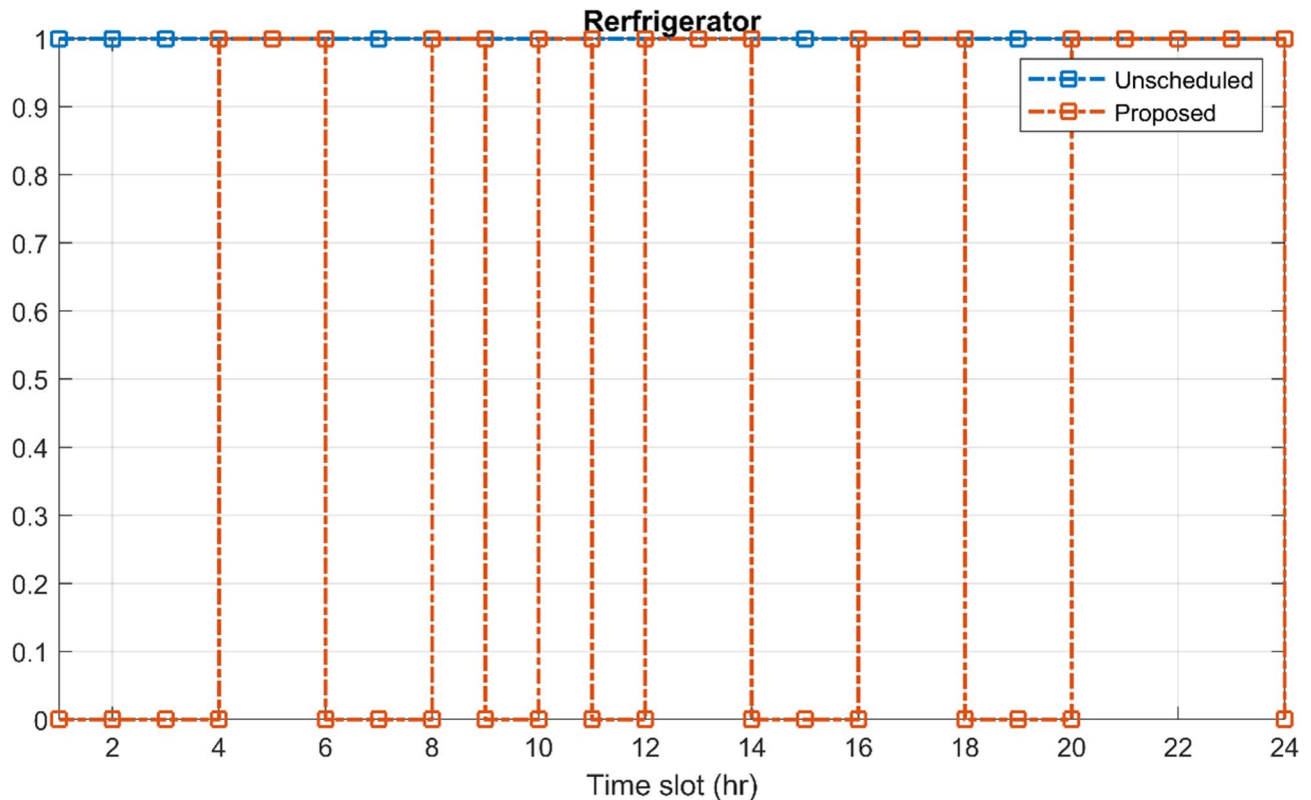


Fig. 11. Comparison of Unscheduled and scheduled patterns for refrigerator.

units, with sporadic reductions in the “Proposed” line to as low as 0.2 units, suggesting potential energy-saving strategies or adjustments in usage patterns.

Figure 12 shows various metrics over 24 h. Plotting the Average Electricity Cost, Chart 1 exhibits daily peaks and valleys ranging from 0 to 8. Regarding the cost of energy, Chart 2, which shows Average User Dissatisfaction, shows an inverse trend with values ranging from 1.9 to 2.4, indicating that higher costs may be associated with higher levels of unhappiness. In the meantime, Chart 3 shows the Average Battery Level, which rises consistently from 50 to 100, signifying a steady charging procedure. The complicated relationships among these metrics are suggested by the connected trajectories of customer discontent and electricity cost, as well as the constant rise in battery level.

Three-line charts with varied y-axis scales that indicate various metrics from separate episodes are displayed in Fig. 13. The first chart, Average Electricity Cost, exhibits peaks and valleys during the episodes, ranging from 0 to 3. In relation to the cost of energy, Chart 2, Average User Dissatisfaction, which spans from 1 to 2.5, shows an inverse pattern, indicating that greater costs can be associated with lower user satisfaction. The Average Battery Level, displayed in Chart 3, indicates an overall rising trend but varies between 20 and 100. While the battery level generally gets better over time despite variations, the dynamic structure of all three figures shows complex relationships, including possible links between electricity cost and customer displeasure.

Figures 14, 15, 16 and 17 provide a 24-hour comparison of power costs and energy use using different approaches. The Unscheduled approach, as depicted in Fig. 13, consistently consumes around 4 kW of energy, but the Proposed (red), MORL-POA (purple), SAPOA (pink), and POA (blue) techniques demonstrate reduced consumption, especially during peak hours (10–16). Figure 14 illustrates that when using Renewable Energy Sources (RES), MORL-POA-With RES displays the lowest peak consumption, whereas the Proposed-With RES technique has a greater peak usage. Figures 15 and 16 show that POA (blue) has the lowest electricity prices, while the Unscheduled approach has the greatest expenditures, especially during peak hours (10–16). Figure 16’s RES integration often lowers expenses; the most cost-effective option is MORL-POA-With RES.

Comparative analysis with state of art techniques

Table 6 compares the Proposed Model to various cutting-edge smart home energy management systems using key criteria such as optimization approach, adaptability, user preference integration, PAR reduction, cost savings, decision-making flexibility, and data utilization. The Proposed Model distinguishes itself by combining Deep Q Networks (DQN) with the Self-Adaptive Puma Optimizer Algorithm (SAPOA), which provides dynamic, continuous learning and real-time user feedback integration, resulting in a significant reduction in PAR as well as optimal energy consumption and appliance scheduling.

Other models, such as those by Islam et al. (2024) and Afroosheh et al. (2024), focus on predictive optimization or energy layout, but do not provide the same level of user preference adaptation and real-time feedback. The

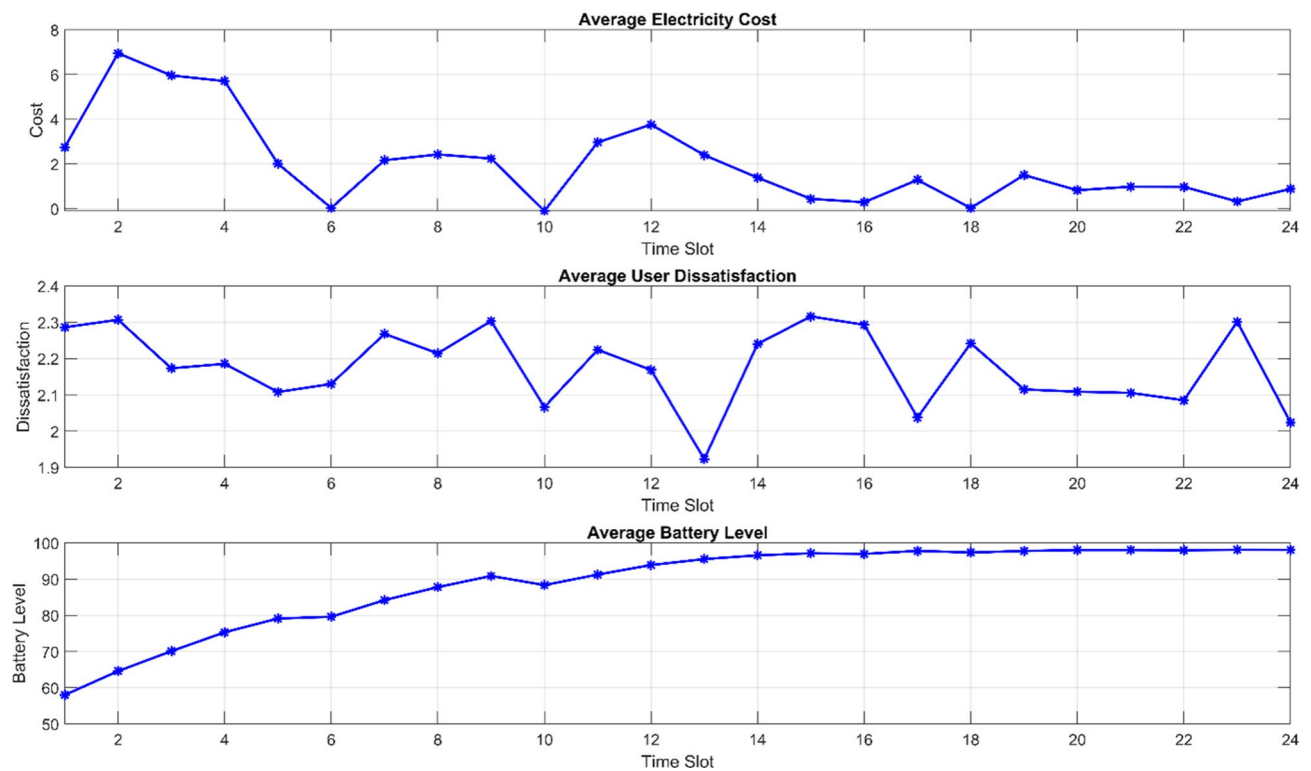


Fig. 12. Graphical representation of the RL based on time.

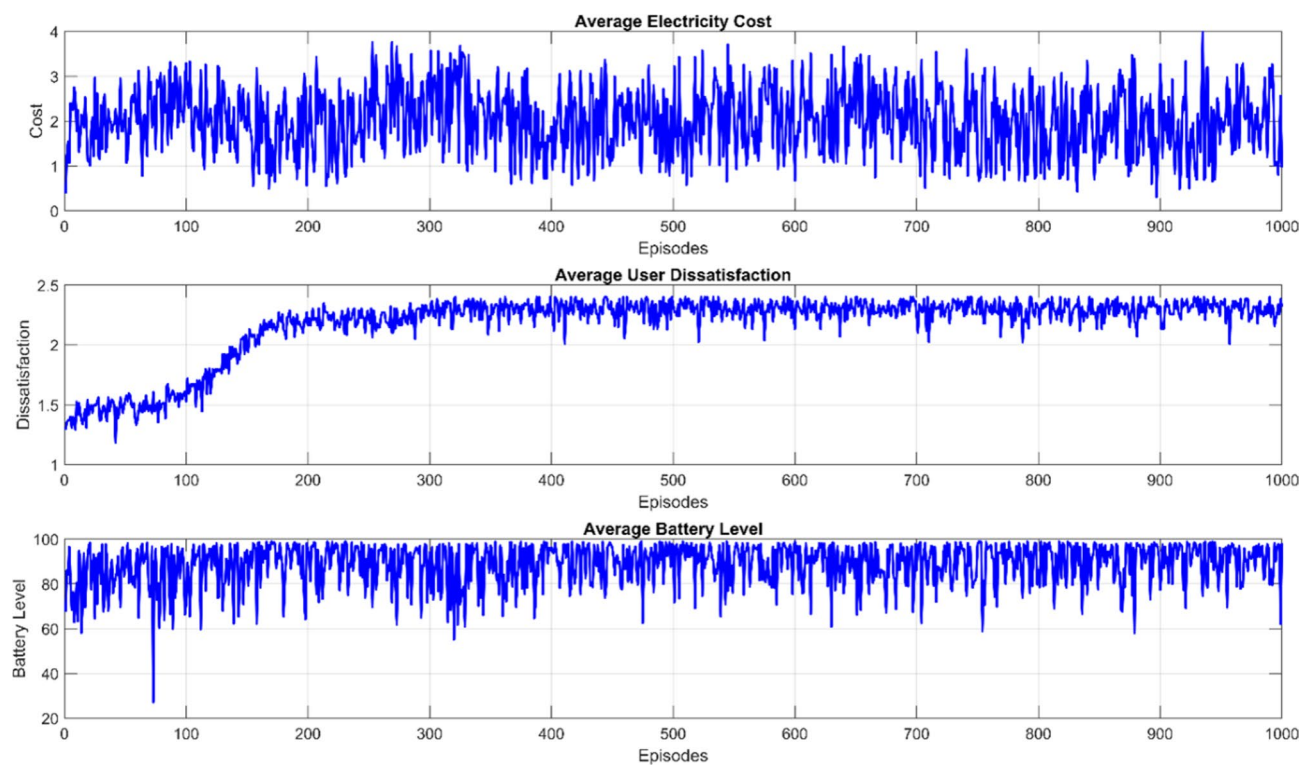


Fig. 13. Graphical representation of the RL based on Episodes.

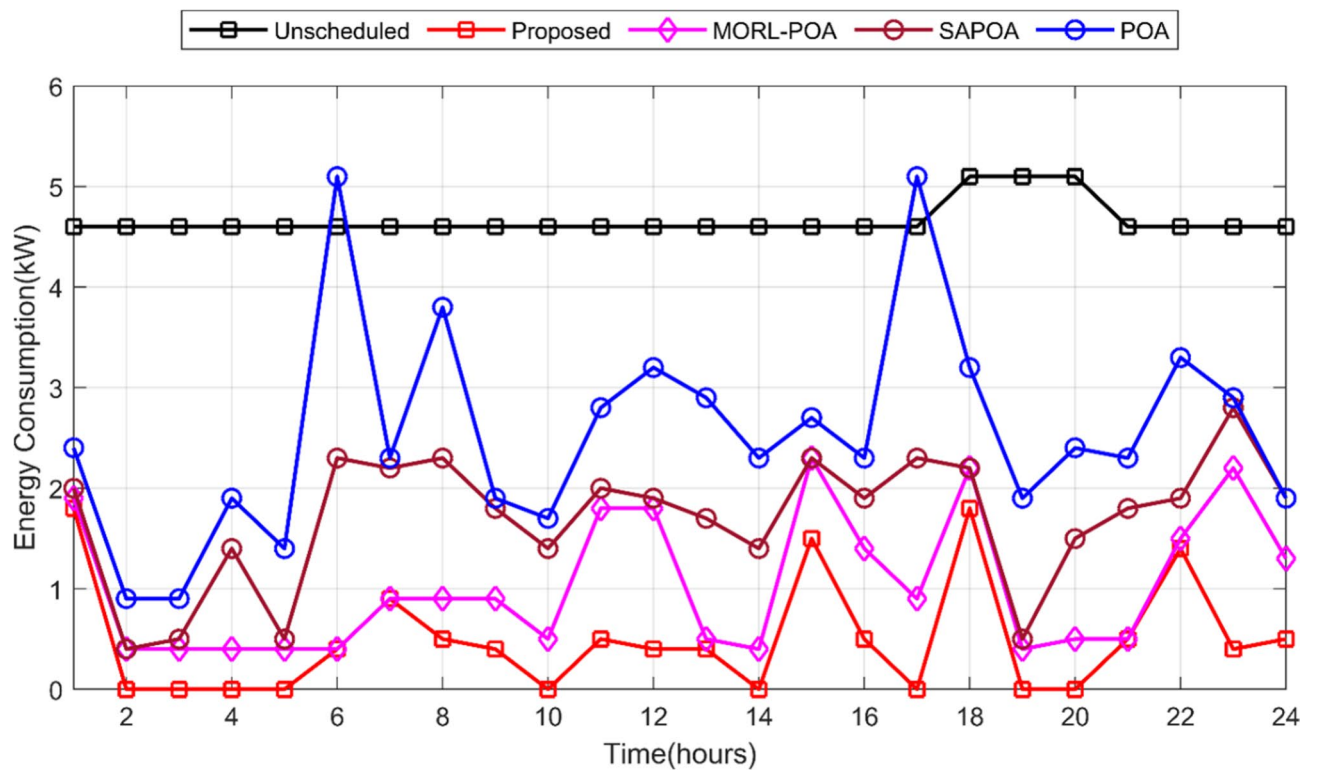


Fig. 14. Comparison of Unscheduled and scheduled patterns of Energy consumption for proposed, MORL-POA, SAPOA, and POA.

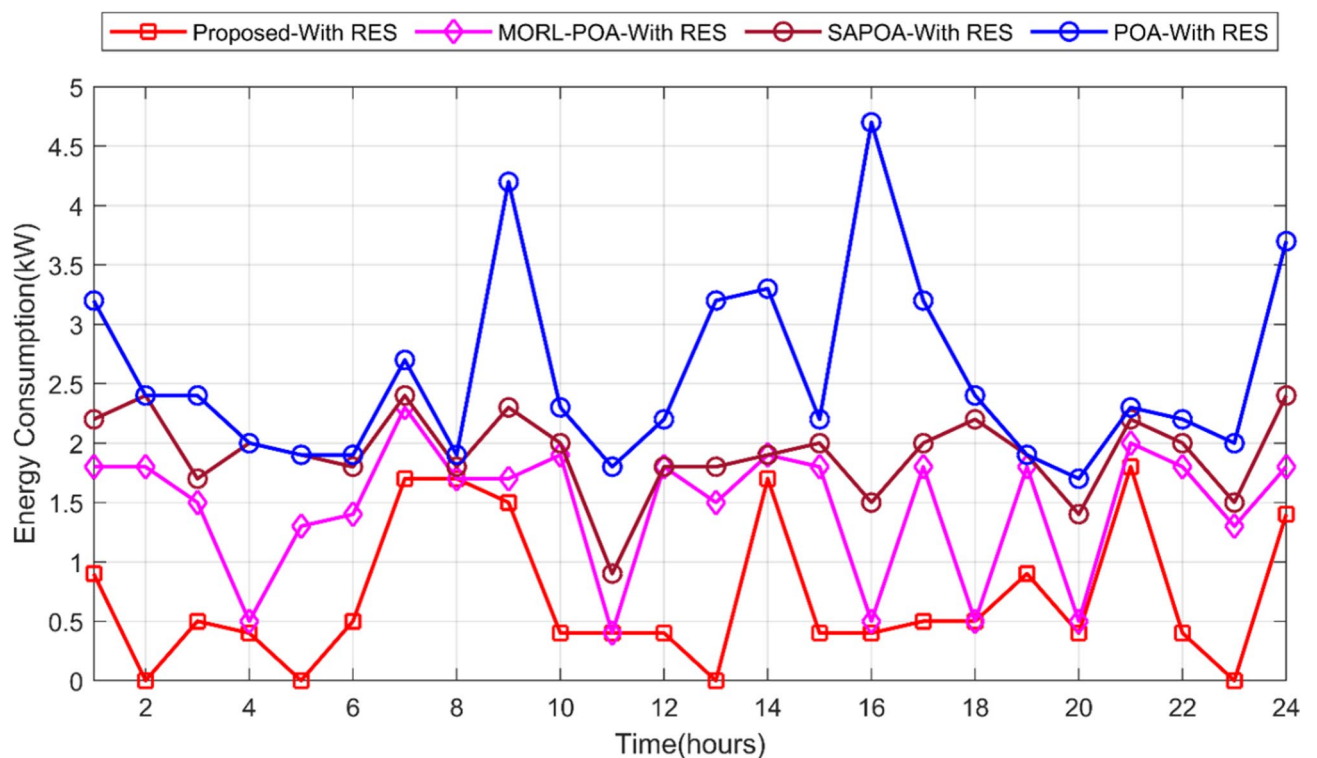


Fig. 15. Comparison of Energy consumption for proposed, MORL-POA, SAPOA, and POA with RES.

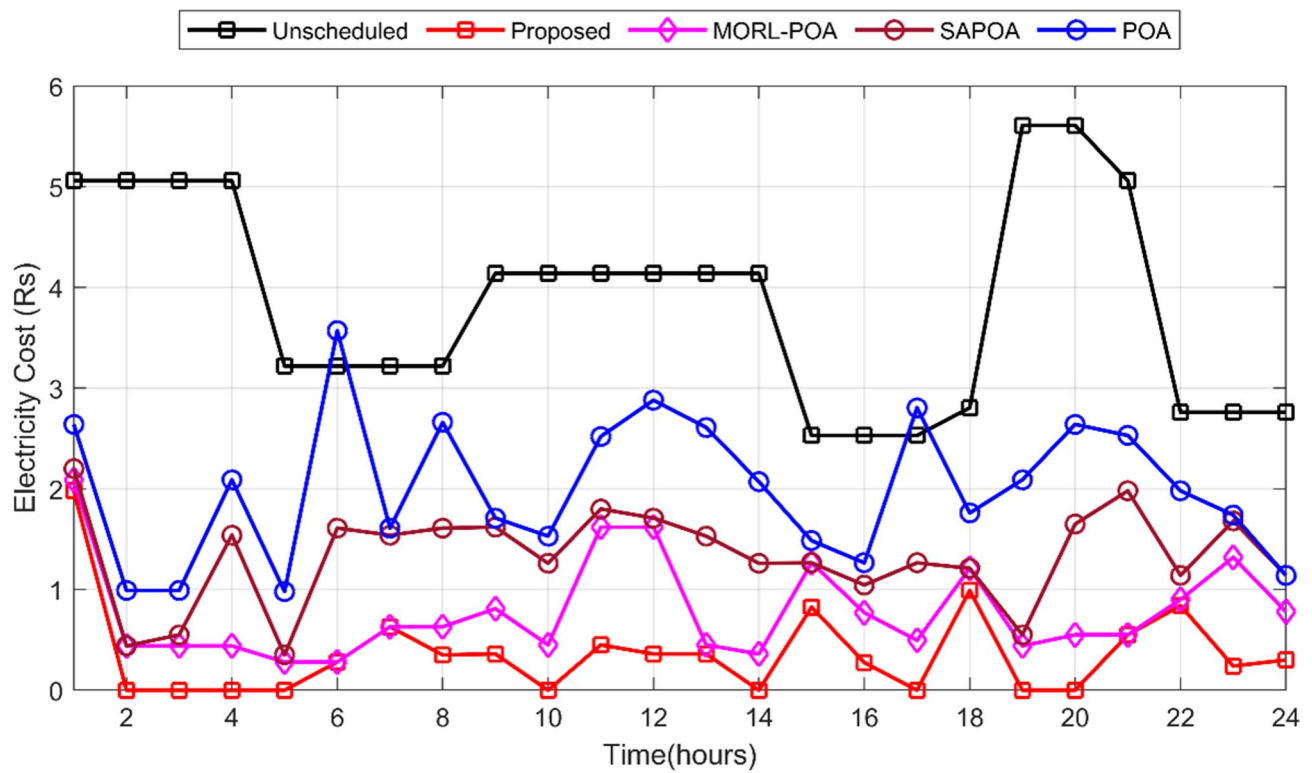


Fig. 16. Comparison of unscheduled and scheduled pattern of electricity cost for proposed, MORL-POA, SAPOA, and POA with RES.

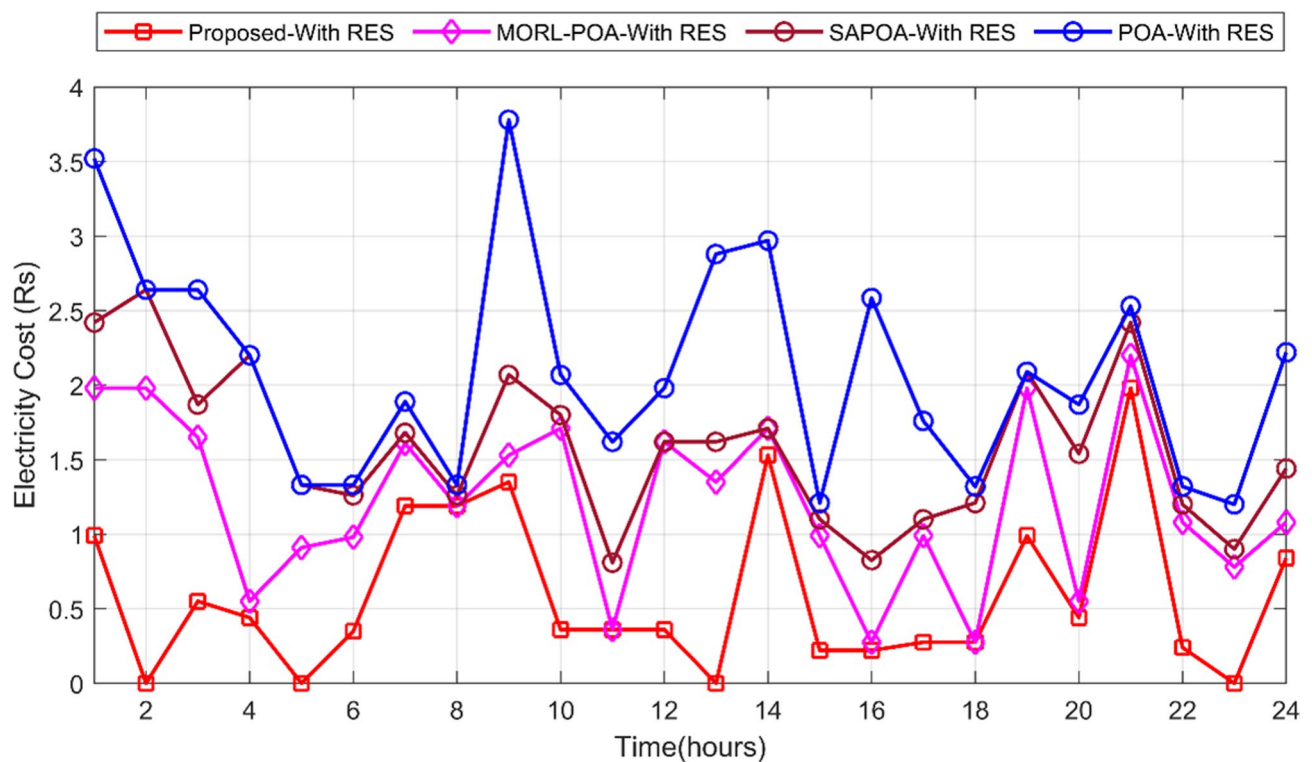


Fig. 17. Comparison of electricity cost for proposed, MORL-POA, SAPOA, and POA with RES.

Method/Criteria	Optimization Approach	Adaptability	User Preference	PAR Reduction	Cost Savings	Decision-making Flexibility	Data Utilization	Experimental Results
Proposed Model	Deep Q Networks, SAPOA, Reinforcement Learning, Historical Data	Dynamic, Continuous Learning	Real-time Feedback integration	1.0339 with RES (from 3.4286 to 1.9765 without RES)	Significant, User-centric	Enhanced flexibility through MO-DQN	Historical Data for Optimization	Dramatic PAR reduction, optimized appliance scheduling
Islam et al., 2024	Machine Learning-based appliance optimization	Adapts to changing conditions	Predictive models, less real-time integration	No specific PAR reduction	Cost-focused, no detailed savings	Real-time optimization but less flexible	Predictive appliance usage models	No experimental results, focuses on appliance optimization
Afroosheh et al., 2024	DRL for optimal energy management	Optimizes energy layout	No real-time user preference integration	No PAR reduction mentioned	Energy layout optimization, no cost details	Less dynamic decision-making	Layout-based optimization	Energy layout optimization, no PAR reduction
Kaewdornhan et al., 2023	Multi-Agent DRL for multi-home energy management	Adapts to multi-home systems	Focuses on EV charging, limited user preference focus	No specific PAR reduction	Multi-home energy management, no user-centric focus	Multi-agent, less flexible decision-making	Multi-home energy management	Results show real-time scheduling but no-PAR reduction
Kanso et al., 2024	Automated energy management framework	Automated adjustments but less dynamic	No deep user feedback	No PAR reduction mentioned	No cost savings details	Limited flexibility due to automation	Basic feedback integration	Focuses on automation, no PAR or cost savings results
Li and Xu, 2024	Hybrid evolutionary and machine learning approach for buildings	Adapts to building-level systems	Less focus on individual preferences	No PAR reduction mentioned	Focused on building energy systems, no cost savings details	Limited decision-making flexibility	Uses building system data for optimization	Results focus on building-level management, no PAR reduction
Ren et al., 2024	DRL-based system optimization with uncertain parameters	Optimizes under uncertainty	No direct user preference integration	No PAR reduction reported	Energy optimized under uncertainty, cost savings not detailed	Based on uncertainty parameters, lacks real-time feedback	Focuses on uncertain parameters	Optimization under uncertainty, no PAR reduction
Saberi et al., 2024	Multi-agent Deep Q-Learning for uncertainty management	Adapts to uncertain environments	Considers preferences, less integration	No PAR reduction value mentioned	Uncertainty-based energy management, no savings details	Multi-agent, less flexible than proposed model	Data used for constrained optimization	Results focus on multi-agent systems, no PAR reduction

Table 6. Performance comparison of the proposed model with other models.

suggested model is also superior in terms of flexibility and performance in uncertain conditions, showing its capacity to handle dynamic situations effectively, which other approaches lack or to a lesser degree.

Discussion

The main innovation of the suggested model is its combination of state-of-the-art reinforcement learning and optimization methods for dynamic smart home energy management with a focus on user preferences. It couples an MO-DQN with SAPOA for adaptive appliance scheduling through real-time feedback and user-centric design. The approach includes user preferences, adapts dynamically to real-time energy consumption and cost, learns by experience, aims at big appliances, and tracks performance indicators. By focusing on energy-guzzling appliances, the model increases energy efficiency, lowers the cost of running, and beats other models. But the Smart Home Energy Management System (HEMS) is constrained by inflexibility, static optimization, complicated decision-making, management of uncertainty, and integration of user feedback, and therefore more sophisticated strategies are required to overcome these challenges.

Results-based objective validation

The research goals were well met through the integration of the SAPOA with an MO-DQN, which was found to enhance energy management significantly. The dynamic adaptability of the system to user needs was achieved through its enhanced scheduling of home appliances, leading to enhanced energy efficiency and reduced costs. This flexibility was also validated by an impressive 42.5% reduction in the PAR from 3.4286 to 1.9765 without renewable energy integration and by 69.8% to 1.0339 with renewable energy integration. These results affirm the ability of the model to balance multiple goals, including energy consumption, cost, and user comfort, hence outperforming traditional methods. The focus on performance-enhancing indicators, such as PAR and energy consumption, is towards demonstrating practical applicability and better performance. The user satisfaction focus of the system was achieved through live learning and dynamic tuning so that the decisions related to energy management were steered according to user expectations, ultimately resulting in greater user comfort and reduced dissatisfaction.

The integration of Deep Reinforcement Learning (DQN) and SAPOA in smart HEMS gives robust optimization but demands high computing power, large memory, and longer training period and is therefore unsuitable for low-end or embedded devices. The disadvantages are not support for deep learning and low real-time response on non-GPU devices. To address this, light-weight substitutes include tabular Q-learning

or SARSA, shallow models like decision trees, simpler optimizers like PSO or DE, and heuristic rules for lesser critical tasks. The combination of TinyML, hierarchical control (device-level rules with central optimization), and offloading computationally heavy tasks to the cloud can help improve scalability and efficiency significantly.

Conclusion

The new smart HEMS, with a DR strategy and a SAPOA and a MO-DQN, effectively addresses the dynamic problems of energy consumption, cost, and user preference. Unlike conventional methods that rely on static optimization and basic reinforcement learning, this new approach dynamically responds to real-time energy prices, user preferences, and load demands, ensuring low electricity bills and maximum energy utilization. The SAPOA optimizes appliance scheduling by real-time adaptation of parameters, while the MO-DQN improves decision-making by learning from past energy consumption patterns. This integration not only controls the uncertainties in forecasting energy expenditures but also coordinates appliance usage with user desires, especially for high-energy appliances such as air conditioners, washing machines, and refrigerators. Experimental results, deployed on the Matlab platform, showed notable improvements in the performance indicators, such as the decrease in the PAR from 3.4286 to 1.9765 without renewable energy and to 1.0339 with renewable energy integration. These outcomes surpass current reinforcement learning and heuristic-based optimization methods, which demonstrate the system's improved adaptability and effectiveness. Overall, this new generation energy management system promises greater efficiency in energy utilization, substantial economic advantages, and greater user contentment, promising more intelligent, responsive home energy solutions.

Future scope

Subsequent study undertakings will explore the capabilities of advanced machine learning techniques coupled with real-time data analytics as a means of further enhancing flexibility and precision with intelligent HEMS. Further enhancements might come from extending the system's use to various energy sources and surroundings and looking into the integration of cutting-edge technologies like blockchain for safe energy transactions. In dynamic energy circumstances, performance and satisfaction with services could be further optimized by including user feedback loops and personalization capabilities. The key improvement areas are as follow.

- **Real-world applicability and IoT integration:** Future research will involve the integration of the proposed energy management system with real-time IoT-based smart home devices. The deployment of the system will occur in a real-world setting utilizing IoT-enabled appliances, including smart energy meters, thermostats, and refrigerators, to gather real-time data on energy consumption and user preferences. The system will undergo testing in real-world conditions to evaluate its adaptability to dynamic user behavior and real-time energy pricing. Real-time data will facilitate dynamic adjustments to appliance schedules. The system's performance will be evaluated, focusing on latency, energy savings, user satisfaction, and overall responsiveness. This will facilitate the connection between theoretical simulations and practical deployment, thereby validating the system's performance in real-world smart home environments.
- **Computational efficiency and hardware constraints:** Reinforcement learning algorithms are recognized for their computational intensity, particularly in smart homes where devices may have resource constraints. Future research will involve a detailed examination of the computational demands of the proposed system. This study will examine lightweight alternatives to improve system efficiency and ensure effective operation on resource-constrained devices, including microcontrollers, smart hubs, and energy meters. We intend to employ techniques such as model pruning, quantization, and federated learning to decrease the computational burden. Model pruning facilitates the elimination of redundant parameters from the neural network, thereby decreasing its size and inference time. Quantization reduces the precision of model weights, thereby accelerating computations and decreasing memory usage. We will examine federated learning to decentralize model training across devices, thereby reducing communication overhead and enhancing privacy while ensuring scalability. The incorporation of these techniques is intended to enhance the system's computational efficiency, making it suitable for real-time deployment in smart home environments characterized by limited hardware resources.
- **Lightweight alternatives:** To address the demand for lightweight alternatives, we will investigate additional optimizations in the model to enable operation on devices with reduced processing capabilities. We will focus on utilizing smaller, simpler models for real-time appliance scheduling and energy management, ensuring high accuracy in decision-making. This study will examine the application of edge computing to transfer substantial computations to proximate devices, thereby reducing latency and enhancing system responsiveness.
- **Integration of advancements in reinforcement learning for smart grids:** To address the gap in the analysis of recent advancements in reinforcement learning-based smart grids, we aim to improve the proposed model by incorporating advanced techniques from multi-agent reinforcement learning (MARL) and decentralized learning. Recent advancements in multi-agent reinforcement learning (MARL) enable smart home appliances to collaborate and optimize energy consumption collectively, leading to enhanced energy efficiency across various devices. The integration of these techniques will enhance the system's adaptability to real-time environmental changes and improve its effectiveness in energy resource management. We will implement real-time demand forecasting to predict energy consumption patterns and adjust appliance scheduling accordingly. This approach allows for proactive management of energy consumption according to anticipated demand, thereby decreasing costs and enhancing efficiency, particularly in residences equipped with renewable energy sources like solar panels. The integration of real-time forecasting will enable the system to accommodate variations in energy supply, demand, and pricing, thereby improving its overall effectiveness.
- **Scalability and user preferences:** Future research will investigate the scalability of the proposed system for larger residential settings or neighborhood-scale energy management. The system will undergo testing with a

progressively larger number of appliances, homes, and devices to evaluate its performance in more complex scenarios. This study will examine the system's capacity to efficiently manage and adapt to dynamic user preferences, especially in multi-user environments. We aim to ensure that the system can accommodate an increasing number of devices by incorporating advanced forecasting models and emphasizing scalability, thereby maintaining optimal energy management.

Data availability

The dataset will be available with the corresponding author based on individual requests.

Received: 13 December 2024; Accepted: 19 June 2025

Published online: 09 July 2025

References

1. Pinthurat, W., Surinkaew, T. & Hredzak, B. An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages. *Renewable Sustain. Energy Rev.* **202** 114648. (2024).
2. Afroosheh, S., Esapour, K., Khorram-Nia, R. & Karimi, M. Reinforcement learning layout-based optimal energy management in smart home: AI-based approach. *IET Generation, Transmission & Distribution*, **18** (15), 2509–2520. (2024).
3. Ren, K. et al. A data-driven DRL-based home energy management system optimization framework considering uncertain household parameters. *Appl. Energy* **355**, 122258. (2024).
4. Abid, M. S. et al. A novel multi-objective optimization based multi-agent deep reinforcement learning approach for microgrid resources planning. *Appl. Energy*, 353, 122029. (2024).
5. Abishu, H. N. et al. May. Multi-agent DRL-based multi-objective demand response optimization for real-time energy management in smart homes. In: *2024 International Wireless Communications and Mobile Computing (IWCMC)* (1210–1217). (IEEE, 2024).
6. Li, W. & Xu, X. A hybrid evolutionary and machine learning approach for smart building: Sustainable building energy management design. *Sustain. Energy Technol. Assess.* **65** 103709. (2024).
7. Islam, S. M. A., Hossain, M. R. & Jubair, M. Intelligent energy management: leveraging an effective machine learning for predictive appliance energy optimization in smart homes. *Eur. J. Artif. Intell. Mach. Learn.* **3** (1), 18–26 (2024).
8. Kanso, H., Noureddine, A. & Exposito, E. An automated energy management framework for smart homes. *J. Ambient Intell. Smart Environ.*, (Preprint), 1–20. (2024).
9. Yang, C., Fan, Z., Lu, W. & Gao, H. Uncertain iterative optimal attitude control method for periodic satellite with reliability constraint. *IEEE Trans. Aerosp. Electron. Syst.* <https://doi.org/10.1109/TAES.2024.3404915> (2024).
10. Saberi, H., Zhang, C. & Dong, Z. Y. A Multi-Agent Deep Constrained Q-Learning Method for Smart Building Energy Management Under Uncertainties (IEEE Transactions on Smart Grid, 2024).
11. Raza, A. et al. Smart home energy management systems: Research challenges and survey. *Sustainability*, **14** (21), 13840. (2024).
12. Kaewdornhan, N., Srithapon, C., Liemthong, R. & Chatthaworn, R. Real-time multi-home energy management with EV charging scheduling using multi-agent deep reinforcement learning optimization. *Energies* **16** (5), 2357. (2023).
13. Cortez, C., Kasis, A., Papadaskalopoulos, D. & Timotheou, S. Demand management for peak-to-average ratio minimization via intraday block pricing. *IEEE Trans. Smart Grid*. **14** (5), 3584–3599. <https://doi.org/10.1109/TSG.2023.3240522> (2023).
14. Real, A. C., Luz, G. P., Sousa, J. M. C., Brito, M. C. & Vieira, S. M. Optimization of a photovoltaic-battery system using deep reinforcement learning and load forecasting. *Energy AI*, **16**, 100347. (2024).
15. Chen, X., Qu, G., Tang, Y., Low, S. & Li, N. Reinforcement learning for Decision-Making and control in power systems. In *Women in Power: Research and Development Advances in Electric Power Systems* (265–285). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-031-29724-3_10. (2023).
16. Shuai, Q., Yin, Y., Huang, S. & Chen, C. Deep reinforcement Learning-Based Real-Time energy management for an integrated Electric–Thermal energy system. *Sustainability* **17** (2), 407. <https://doi.org/10.3390/su17020407> (2025).
17. You, X., Zhang, P., Liu, M., Lin, L. & Li, S. Epsilon-Greedy-Based MQTT QoS mode selection and power control algorithm for power distribution IoT. *Int. J. Mob. Comput. Multim. Commun.* **14**, 1–18. <https://doi.org/10.4018/ijmcmc.306976> (2023).
18. Sankarananth, S., Karthiga, M., Suganya, E., Sountharajan, S. & Bavirisetti, D. P. AI-enabled metaheuristic optimization for predictive management of renewable energy production in smart grids. *Energy Rep.* **10**, 1299–1312 (2023).
19. Mahapatra, B. & Nayyar, A. Home energy management system (HEMS): concept, architecture, infrastructure, challenges and energy management schemes. *Energ. Syst.* **13** (3), 643–669 (2022).
20. Maurya, P., Tiwari, P. & Pratap, A. Puma optimizer technique for optimal planning of different types of distributed generation units in radial distribution network considering different load models. *Electrical Eng.*, 1–52. (2024).
21. Mir, U., Abbasi, U., Mir, T., Kanwal, S. & Alamri, S. Energy management in smart buildings and homes: current approaches, a hypothetical solution, and open issues and challenges. *IEEE Access*. **9**, 94132–94148 (2021).
22. Nastić, F. et al. Water and electricity consumption and carbon footprint of washing machines. *Energija Ekonomija Ekologija*. <https://doi.org/10.46793/eee23-1.10n> (2023).
23. Forootani, A., Rastegar, M. & Jooshaki, M. An advanced satisfaction-based home energy management system using deep reinforcement learning. *IEEE Access*. (1–1). <https://doi.org/10.1109/ACCESS.2022.3172327> (2022).
24. Blonsky, M., McKenna, K., Maguire, J. & Vincent, T. Home energy management under realistic and uncertain conditions: A comparison of heuristic, deterministic, and stochastic control methods. *Appl. Energy*, 325, 119770. (2022).
25. Abdollahzadeh, B. et al. Puma optimizer (PO): A novel metaheuristic optimization algorithm and its application in machine learning. *Cluster Computing*, 1–49. (2024).
26. Zhang, Z., Wu, Z., Zhang, H. & Wang, J. Meta-learning-based deep reinforcement learning for multiobjective optimization problems. *IEEE Trans. Neural Networks Learn. Systems* (2022).
27. Alabdullah, M. H. & Abido, M. A. Microgrid energy management using deep Q-network reinforcement learning. *Alexandria Eng. J.* **61** (11), 9069–9078. <https://doi.org/10.1016/j.aej.2022.02.042> (2022).
28. Chen, Z. et al. Multi-objective residential load scheduling approach for demand response in smart grid. *Sustain. Cities Soc.* **76**, 103530. <https://doi.org/10.1016/j.scs.2021.103530> (2022).
29. Amer, A. A., Shaban, K. & Massoud, A. M. DRL-HEMS: deep reinforcement learning agent for demand response in home energy management systems considering customers and operators perspectives. *IEEE Trans. Smart Grid*. **14** (1), 239–250 (2022).
30. Sabbioni, L., Daire, L., Bisi, L., Metelli, A. & Restelli, M. Simultaneously updating all persistence values in reinforcement learning. *ArXiv* <https://doi.org/10.48550/arXiv.2211.11620> (2022). abs/2211.11620.
31. Al-Ani, O. & Das, S. Reinforcement learning: theory and applications in Hems. *Energies* **15** (17), 6392. <https://doi.org/10.3390/en15176392> (2022).
32. Aliero, M. S., Qureshi, K. N., Pasha, M. F. & Jeon, G. Smart home energy management systems in internet of things networks for green cities demands and services. *Environ. Technol. Innovation*, **22**, 101443. (2021).

33. Alatawi, M. N. Optimization of home energy management systems in smart cities using bacterial foraging algorithm and deep reinforcement learning for enhanced renewable energy integration. *Int. Trans. Electr. Energy Syst.* **2024** (1), 2194986. <https://doi.org/10.1155/2024/2194986> (2024).
34. Aldahmashi, J. & Ma, X. Real-time energy management in smart homes through deep reinforcement learning. *IEEE Access.* <https://doi.org/10.1109/access.2024.3375771> (2024).
35. Liu, H., Sun, H., Mo, H. & Liu, J. Analysis and modeling of air conditioner usage behavior in residential buildings using monitoring data during hot and humid season. *Energy Build.*, **250**, 111297. <https://doi.org/10.1016/j.enbuild.2021.111297>. (2021).
36. Almughram, O., Abdullah ben Slama, S. & Zafar, B. A. A reinforcement learning approach for integrating an intelligent home energy management system with a vehicle-to-home unit. *Appl. Sci.*, **13**(9), 5539. (2023).

Acknowledgements

The authors extend their appreciation to Taif University, Saudi Arabia, for supporting this work through project number (TU-DSPP-2024-210).

Author contributions

The authors of this work have made significant contributions to its success. Poonam Saroha and Gopal Singh contributed to the foundational research and analysis. Umesh Kumar Lilhore led the project and coordinated the efforts, while Sarita Simaiya played a key role in data interpretation and manuscript preparation. Roobaea Alroobaea, Majed Alsafyani and MD Monish provided expertise in experimental design and technical validation and also helps in revision. Hamed Alsufyani contributed to the development of methodologies and final review. Gopal Singh Guided this research. The collaborative efforts of all authors were instrumental in achieving the outcomes of this research.

Funding

This research was funded by Taif University, Saudi Arabia, Project number (TU-DSPP-2024-210).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025